Foundations of Deep Learning

ALF

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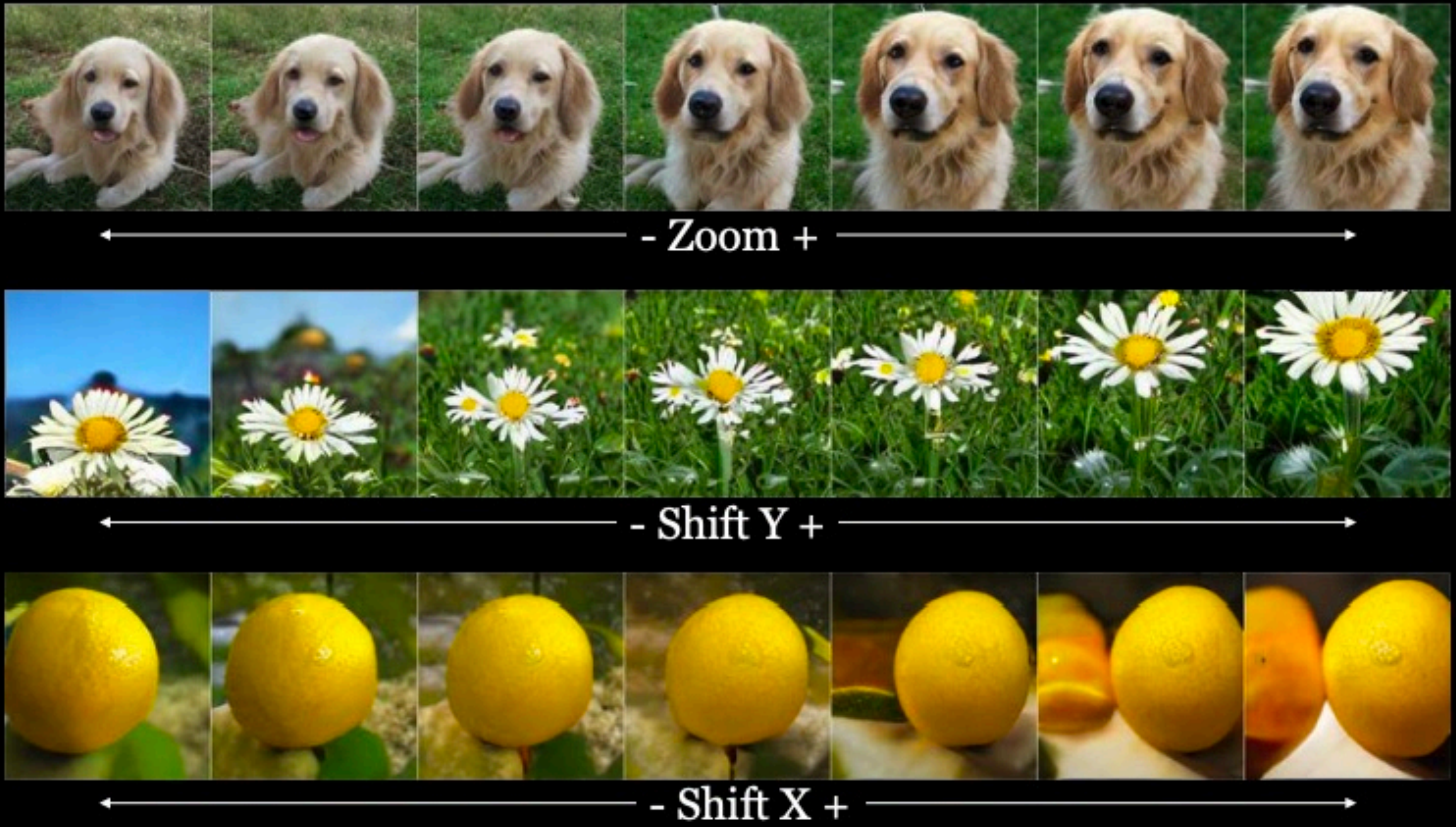
@alfcnz

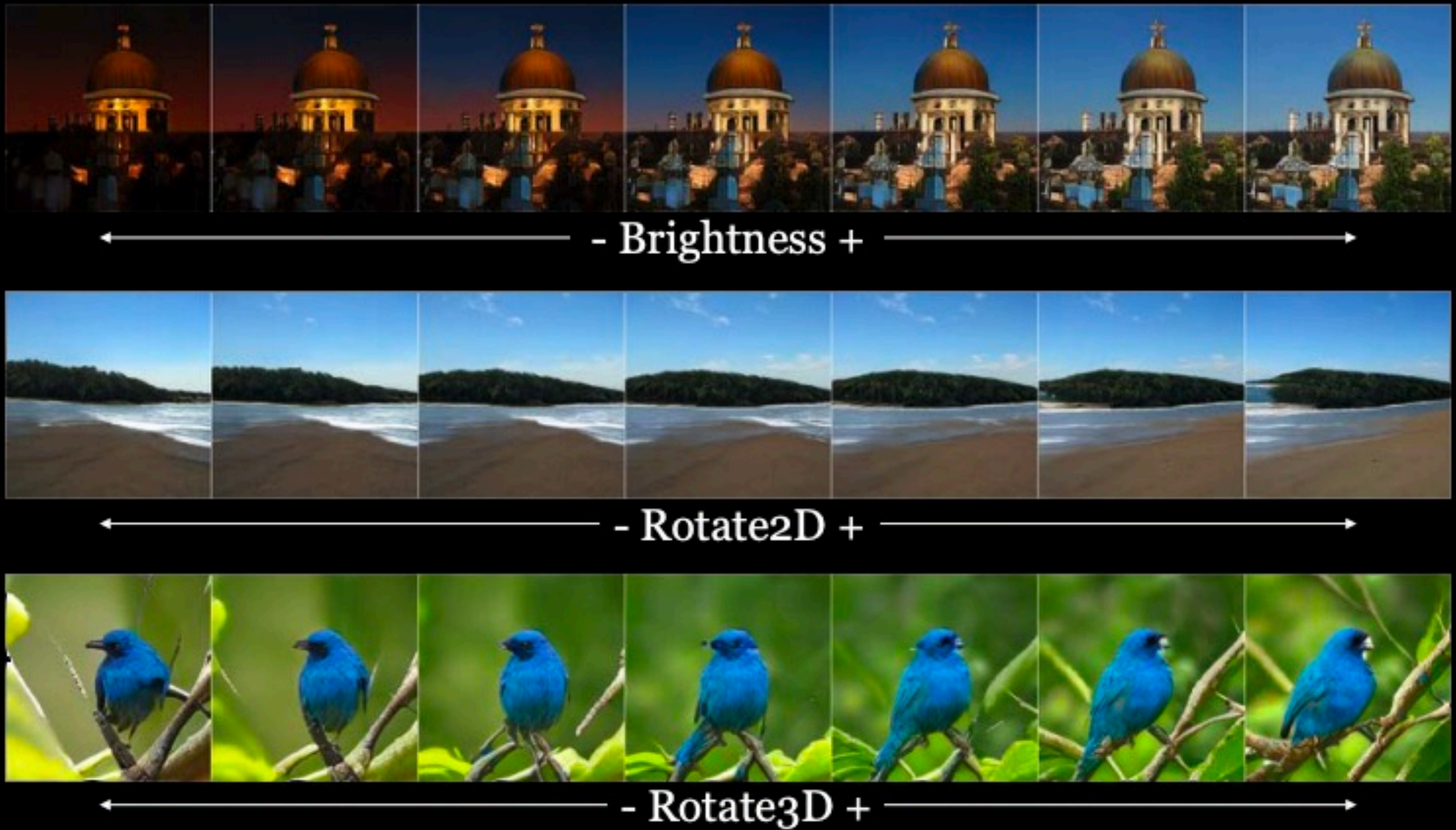
(Self-) Un-supervised learning

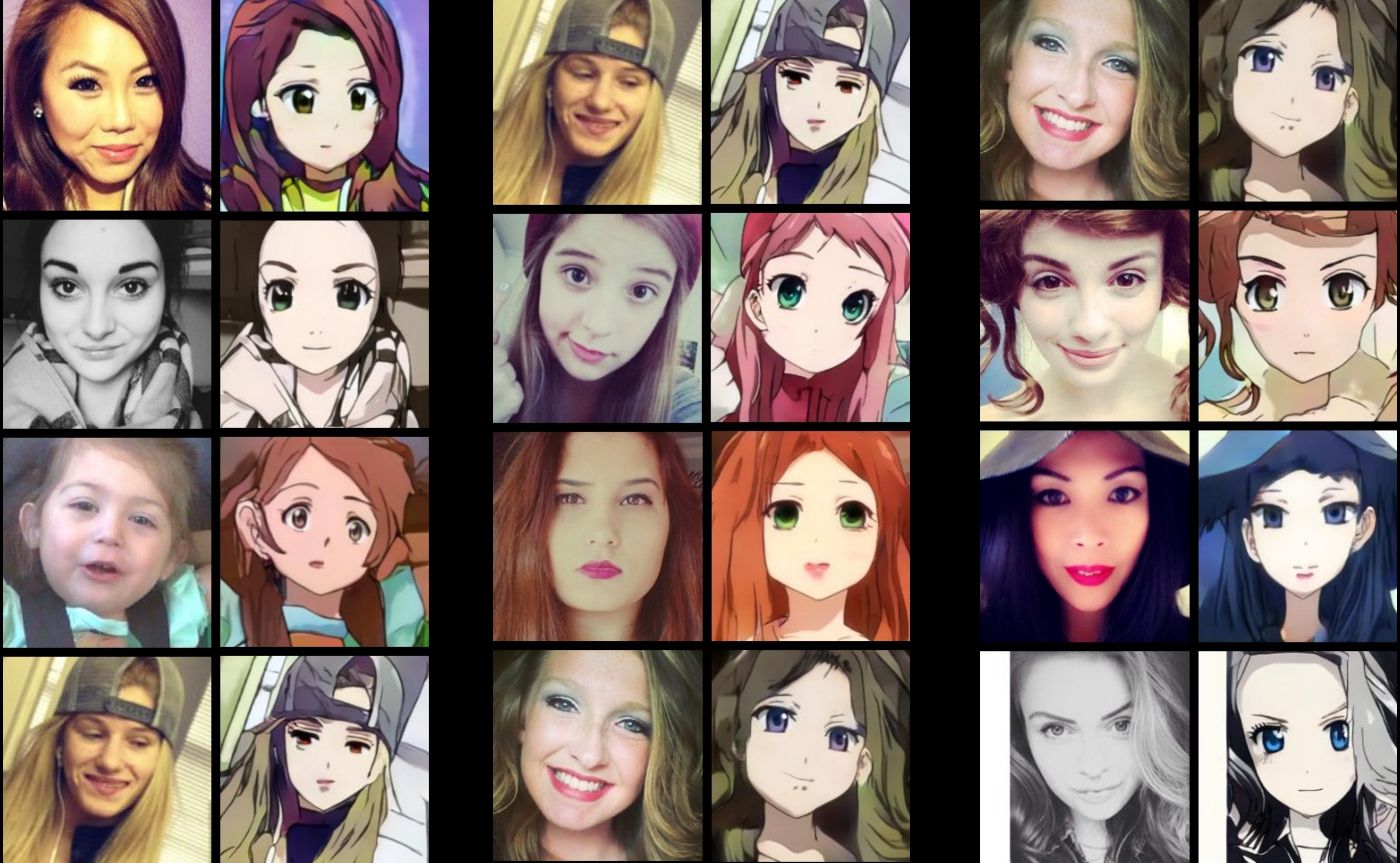
Generative models capabilities

Karras (2019) StyleGAN2 – Analysing and Improving the Image Quality of StyleGAN

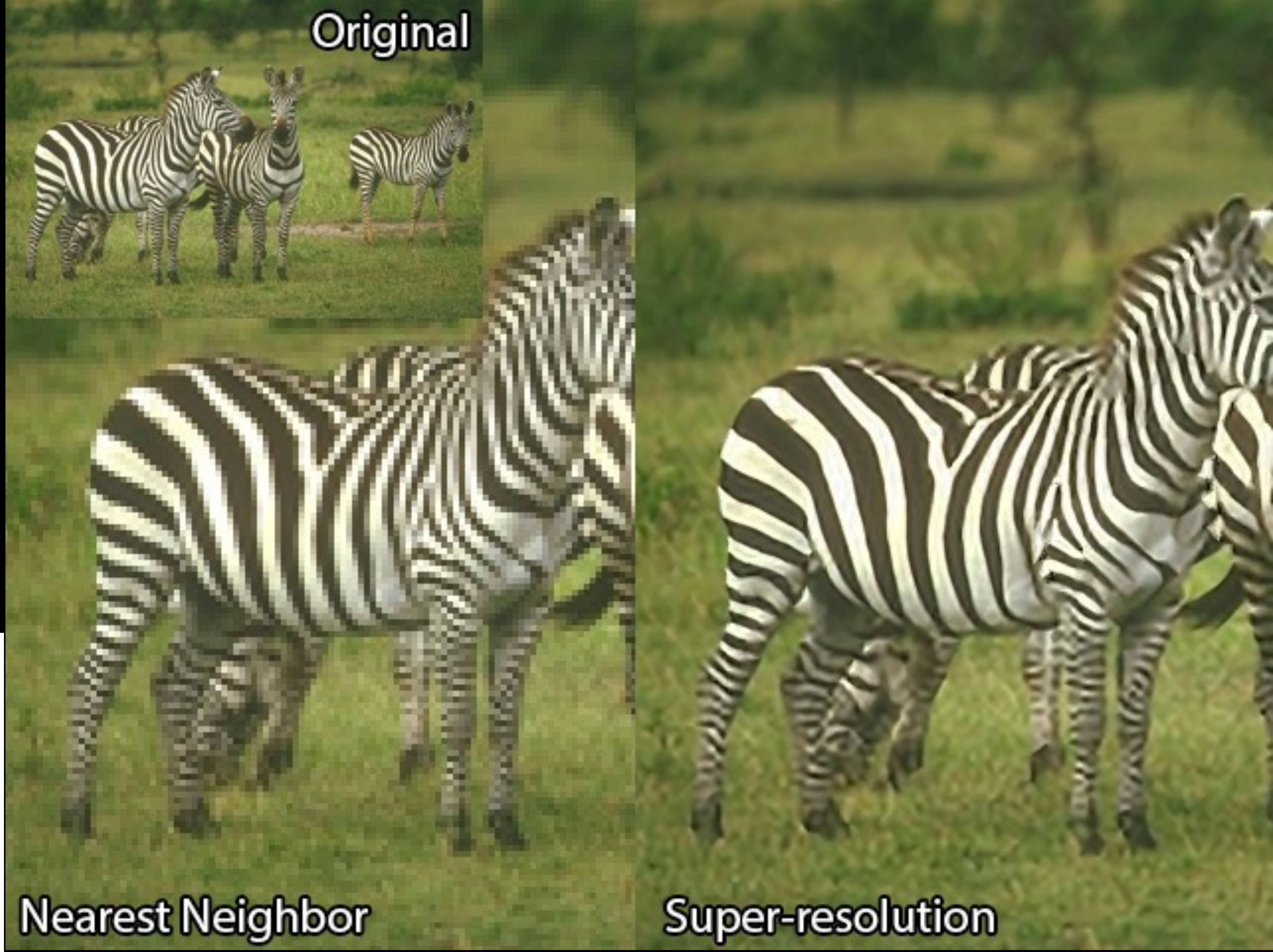
Brock (2018) Large Scale GAN Training for High Fidelity Natural Image Synthesis


- Zoom +
- Shift Y +
- Shift X +


- Brightness +
- Rotate2D +
- Rotate3D +



Super resolution



Glasner (2009) Super-resolution from a single image
www.wisdom.weizmann.ac.il/~vision/SingleImageSR.html

Garcia (2016) srez --- github.com/david-gpu/srez

Inpainting

VAE

GAN

Yeh (2017)
Semantic Image Inpainting
with Deep Generative Models

bit.ly/DCGAN-inpainting

Caption to image

DALL-E

Ramesh (2021) Zero-Shot Text-to-Image Generation

TEXT PROMPT

an armchair in the shape of an avocado. an armchair imitating an avocado.

AI-GENERATED IMAGES

TEXT PROMPT a clock in the shape of an avocado. a clock imitating an avocado.

AI-GENERATED IMAGES

TEXT PROMPT a lamp in the shape of an avocado. a lamp imitating an avocado.

AI-GENERATED IMAGES

TEXT PROMPT a lamp in the form of an avocado. a lamp imitating an avocado.

AI-GENERATED IMAGES

TEXT PROMPT a lamp in the form of a pig. a lamp imitating a pig.

AI-GENERATED IMAGES

TEXT PROMPT

a lamp in the form of a lotus root. a lamp imitating a lotus root.

AI-GENERATED IMAGES

Definitions

x, y, z

Variables' name

Input

x observed during: training ✅ — testing ✅

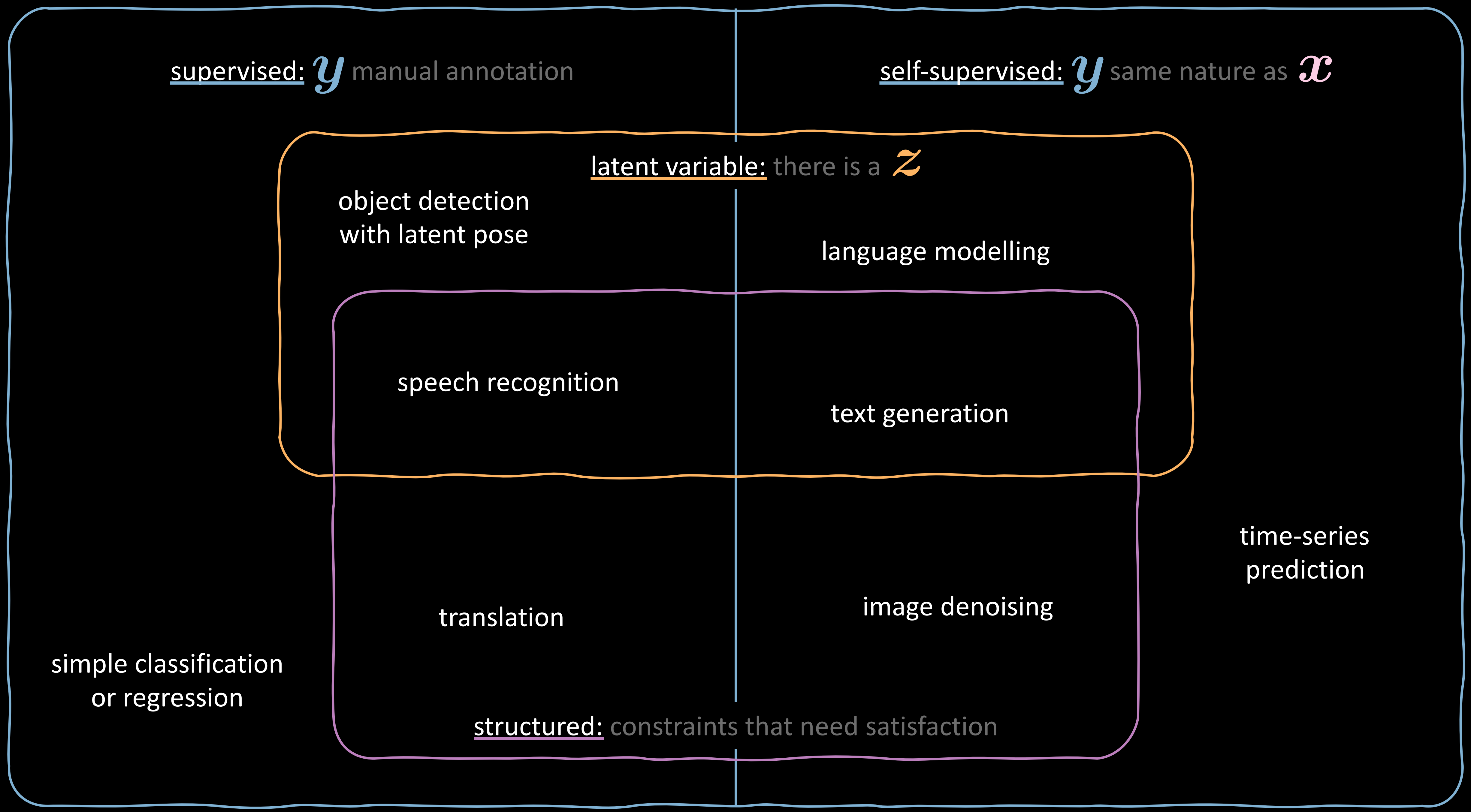
y observed during: training ✅ — testing ❌

z observed during: training ❌ — testing ❌

Output

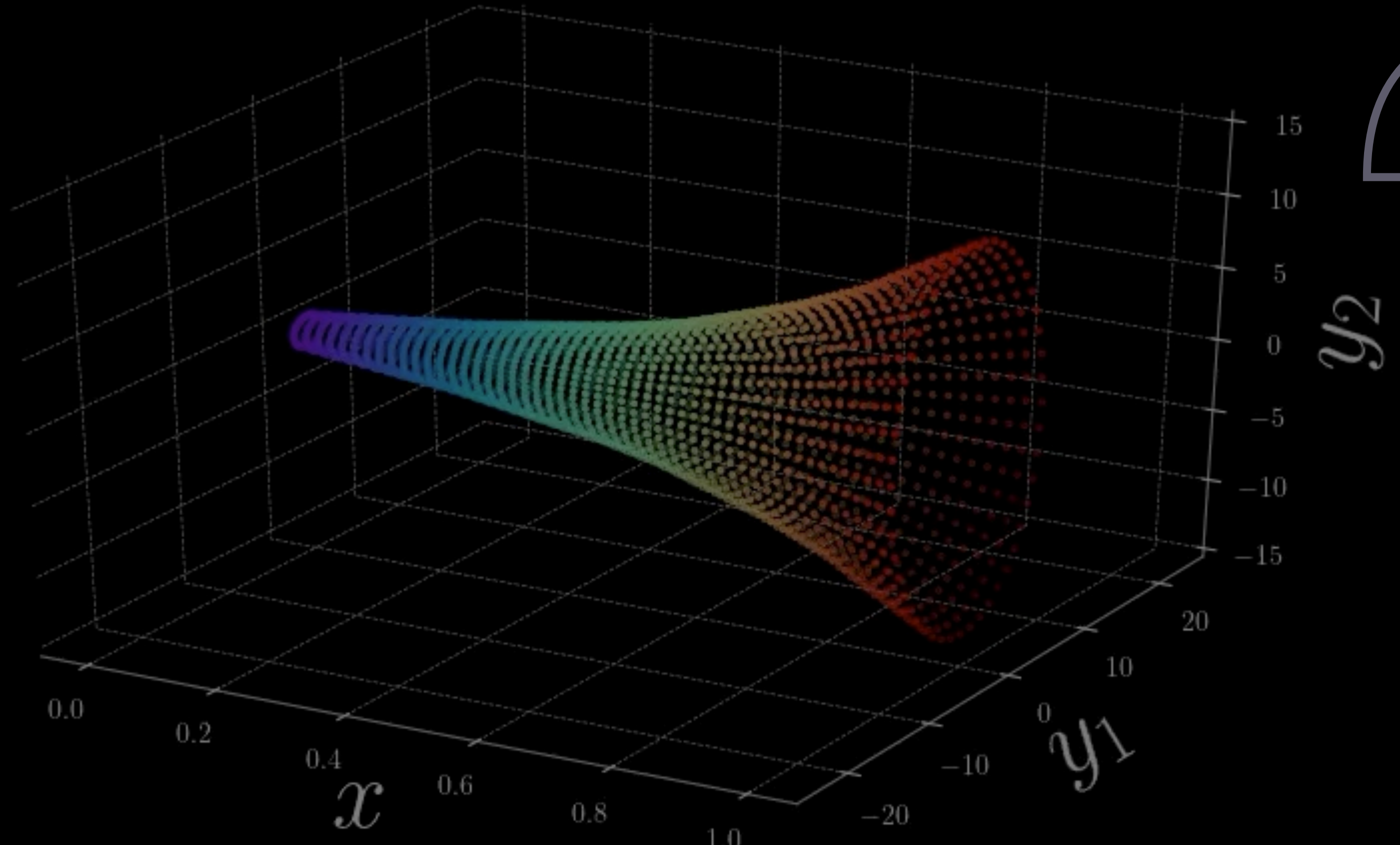
h computed from the input (hidden / internal)

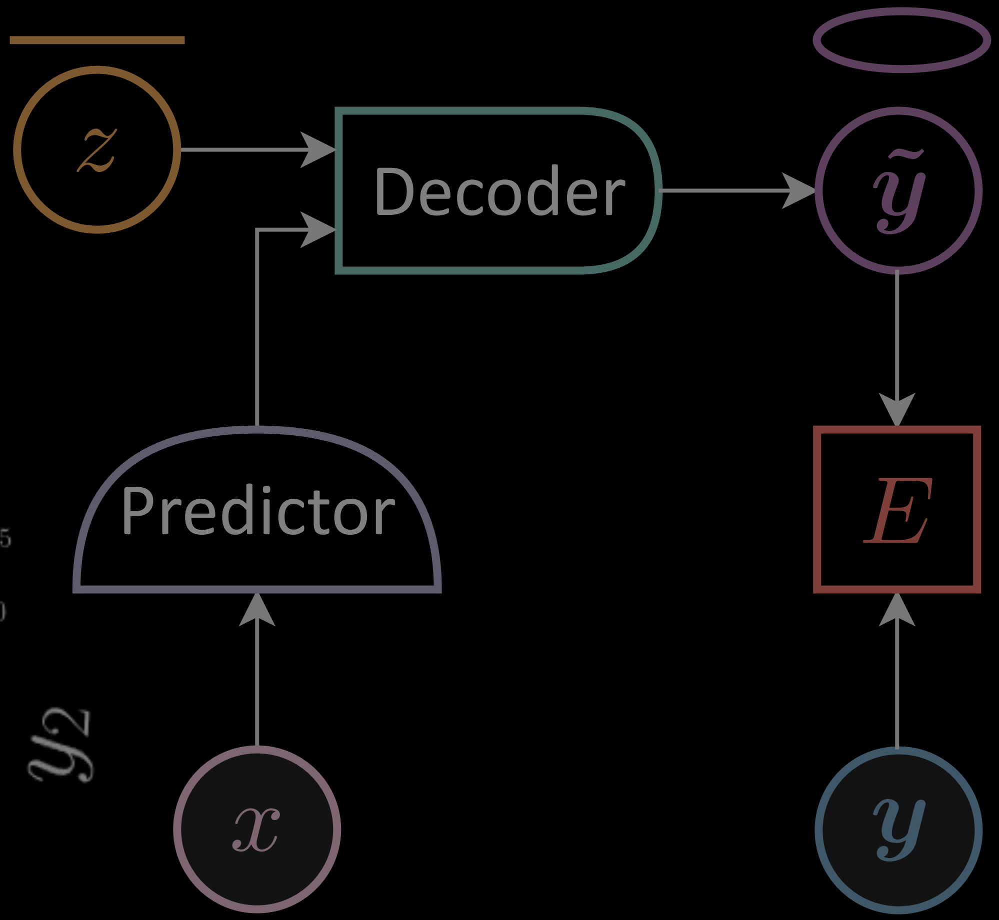
$\tilde{y}$ computed from the hidden (predicted y, ~ means *circa*)


supervised: y manual annotation
self-supervised: y same nature as x
latent variable: there is a z
object detection with latent pose
language modelling
speech recognition
text generation
time-series prediction
translation
image denoising
simple classification or regression
structured: constraints that need satisfaction

from the *EBM* lecture

Trained model manifold



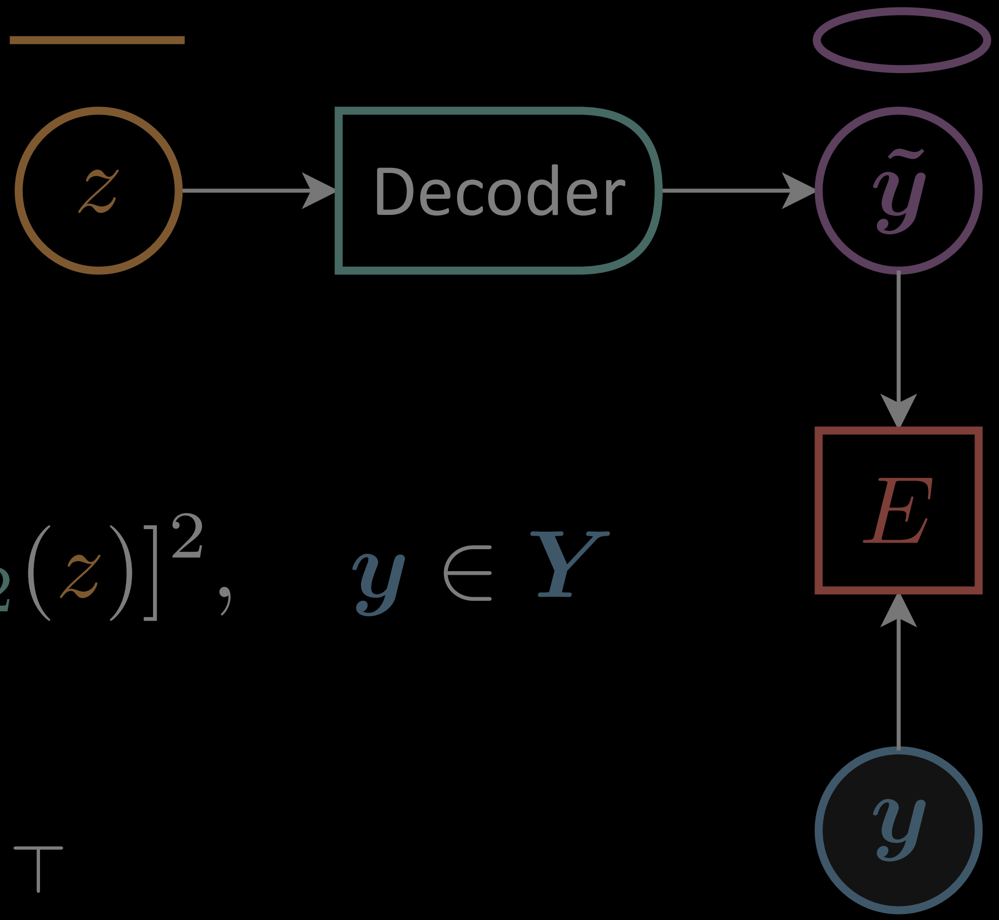


$$z = \left[0 : \frac{\pi}{24} : 2\pi\right[$$

$$x = \left[0 : \frac{1}{50} : 1\right]$$

from the *EBM* lecture

Energy function $E(\boldsymbol{y}, z)$

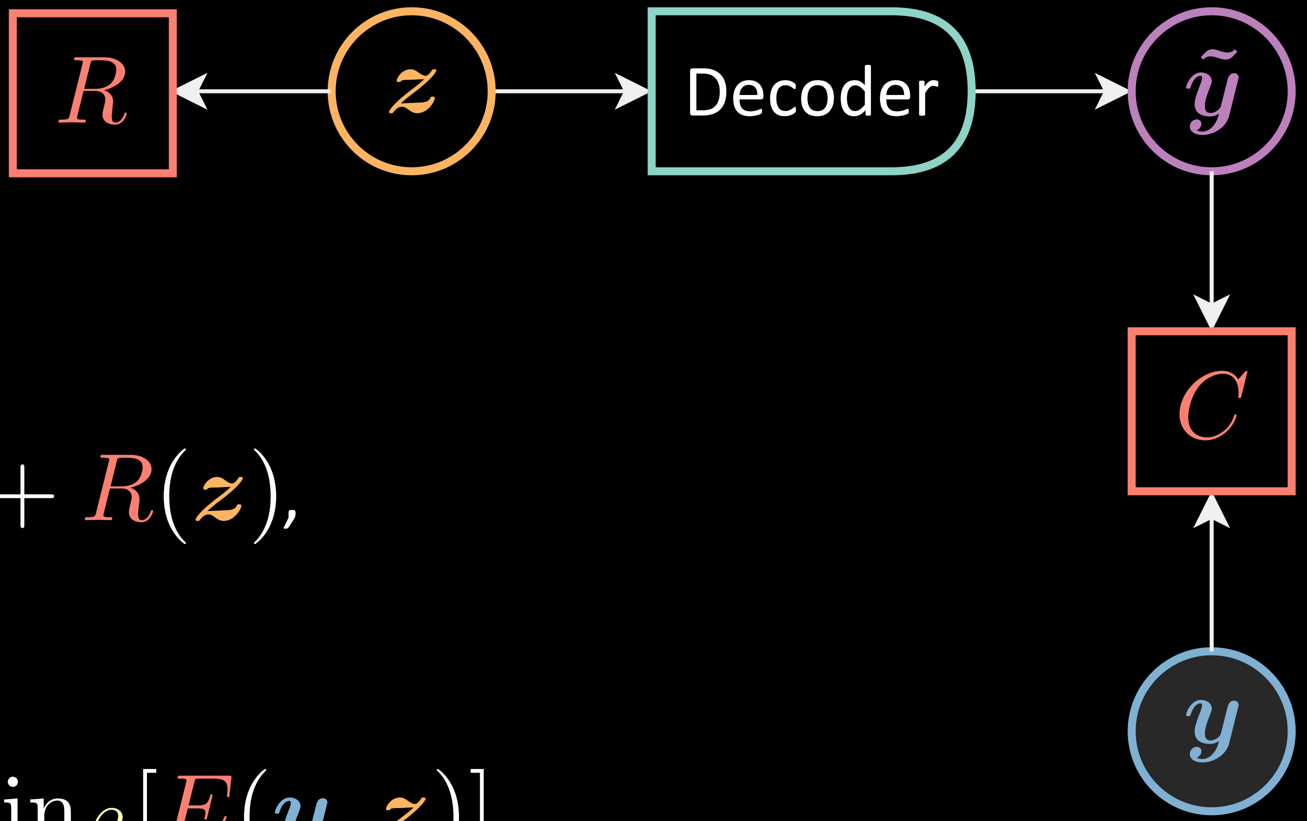


$$E(\boldsymbol{y}, z) = [y_1 - g_1(z)]^2 + [y_2 - g_2(z)]^2, \quad \boldsymbol{y} \in \boldsymbol{Y}$$

$$g = \begin{bmatrix} g_1 & g_2 \end{bmatrix}^\top : \mathbb{R} \to \mathbb{R}^2$$

$$z \mapsto \begin{bmatrix} w_1 \cos(z) & w_2 \sin(z) \end{bmatrix}^\top$$

Training recap



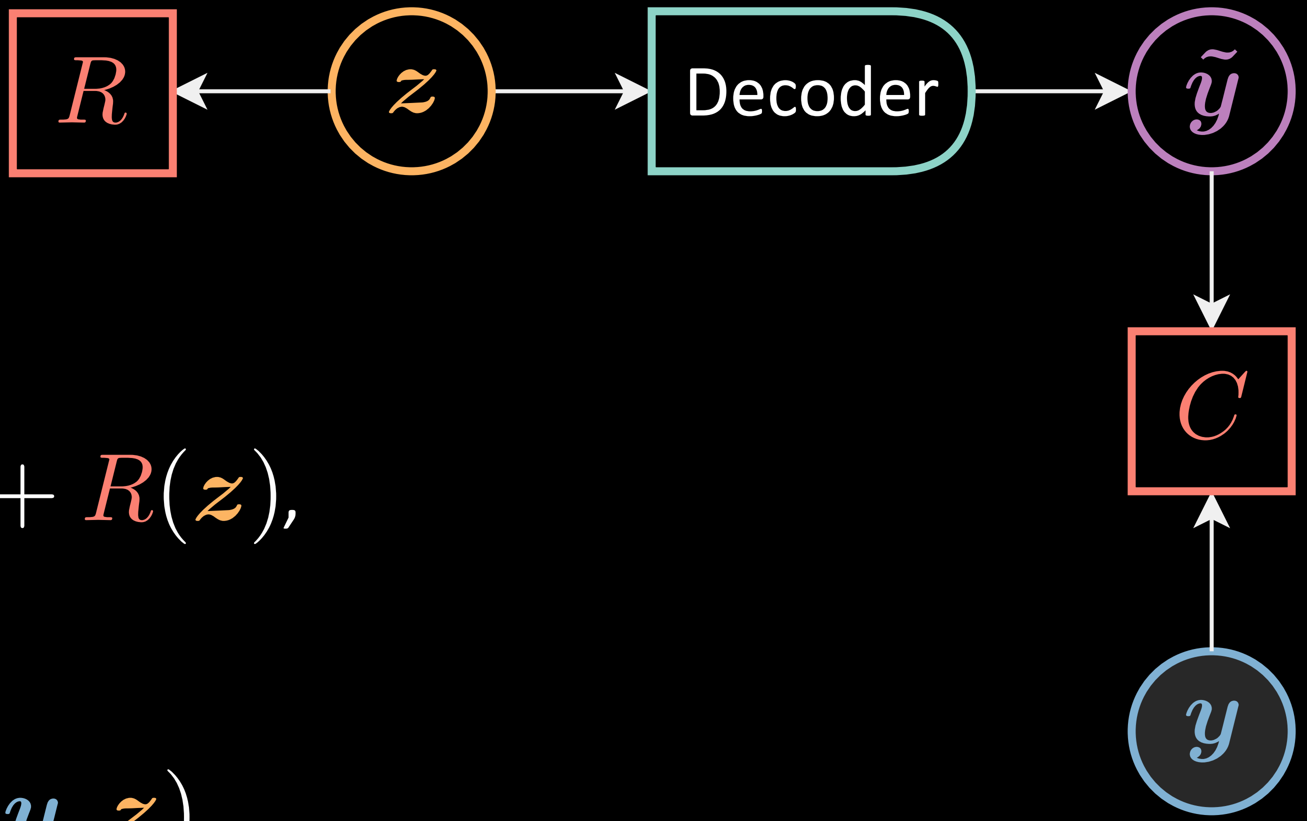
Given an observation y,

given $E(y, z) = C(y, \tilde{y}) + R(z)$,

where $\tilde{y} = \mathrm{Dec}(z)$

- Compute $F_\beta(y) = \underset{z}{\mathrm{softmin}_\beta}[E(y, z)]$
- Minimise $\mathcal{L}(F_\beta(\,\cdot\,), Y)$

Zero temp. limit



Given an observation y,

given $E(y, z) = C(y, \tilde{y}) + R(z)$,

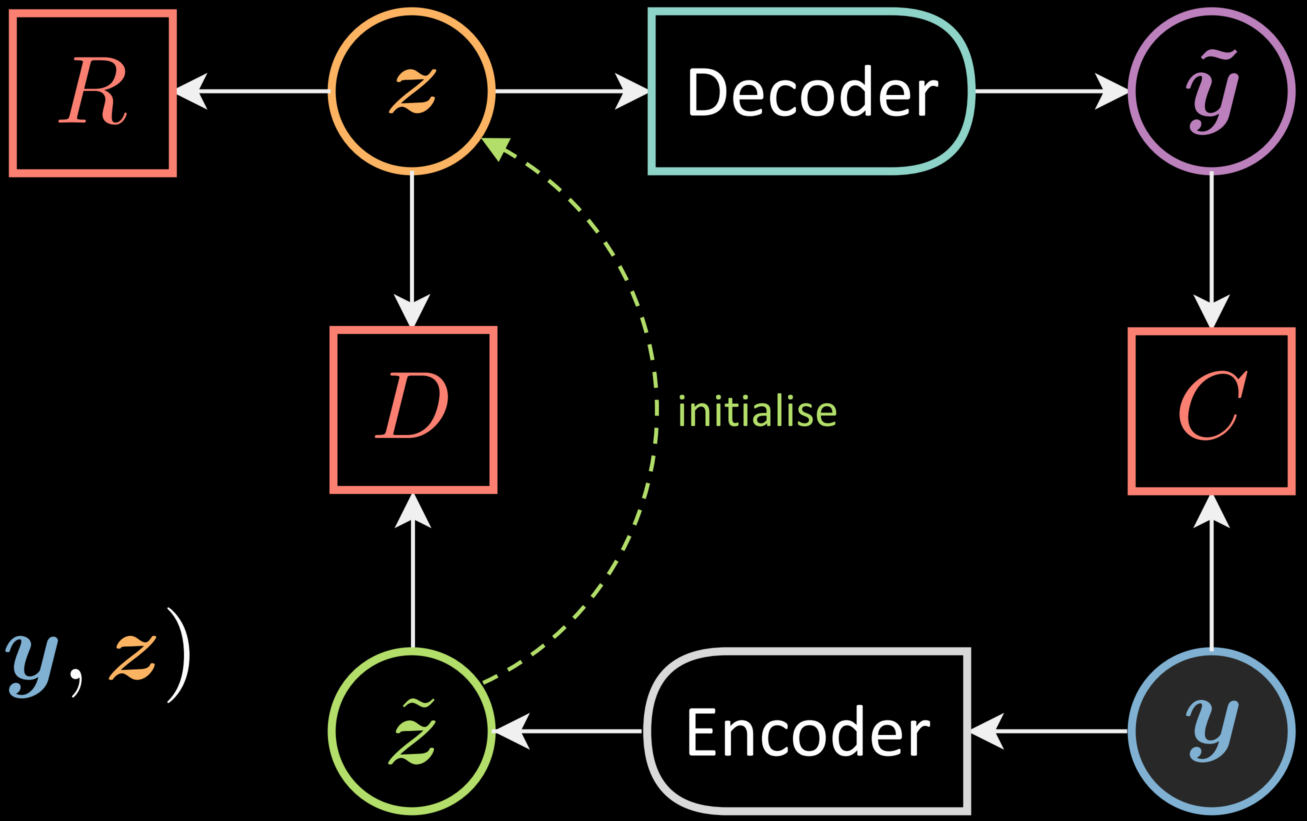
where $\tilde{y} = \mathrm{Dec}(z)$

- Compute $\check{z} = \arg\min_z E(y, z)$

$$F_\infty(y) \overset{z}{=} \min_z E(y, z) = E(y, \check{z})$$

- Minimise $\mathcal{L}(F_\infty(\,\cdot\,), Y)$

Target prop(agation)



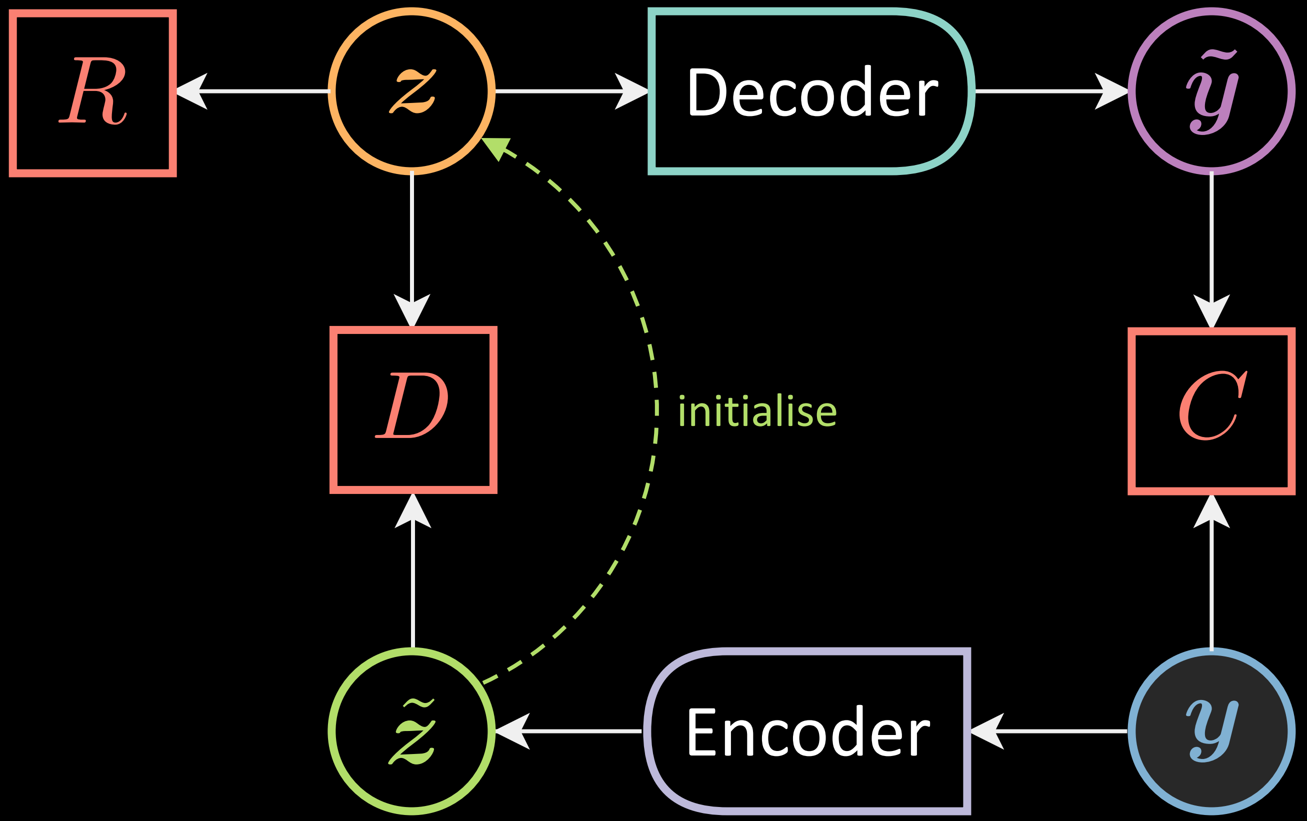
Given an observation y,

- Compute $\tilde{z} = \mathrm{Enc}(y)$
- Compute $\check{z} = \arg\min_{z} E(y, z)$
- Minimise $\mathcal{L}(F_\infty(\cdot), Y)$:

$$\Theta_{\mathrm{Dec}} \leftarrow \Theta_{\mathrm{Dec}} - \nabla_{\Theta_{\mathrm{Dec}}} C(y, \tilde{y})$$

$$\Theta_{\mathrm{Enc}} \leftarrow \Theta_{\mathrm{Enc}} - \nabla_{\Theta_{\mathrm{Enc}}} D(\check{z}, \tilde{z})$$

$$E(y, z) = C(y, \tilde{y}) + R(z) + D(z, \tilde{z})$$



R
z
Decoder
$\tilde{y}$
D
initialise
C
$\tilde{z}$
Encoder
y

Autoencoder

$$\boldsymbol{h} = f(\boldsymbol{W_h y} + \boldsymbol{b_h})$$

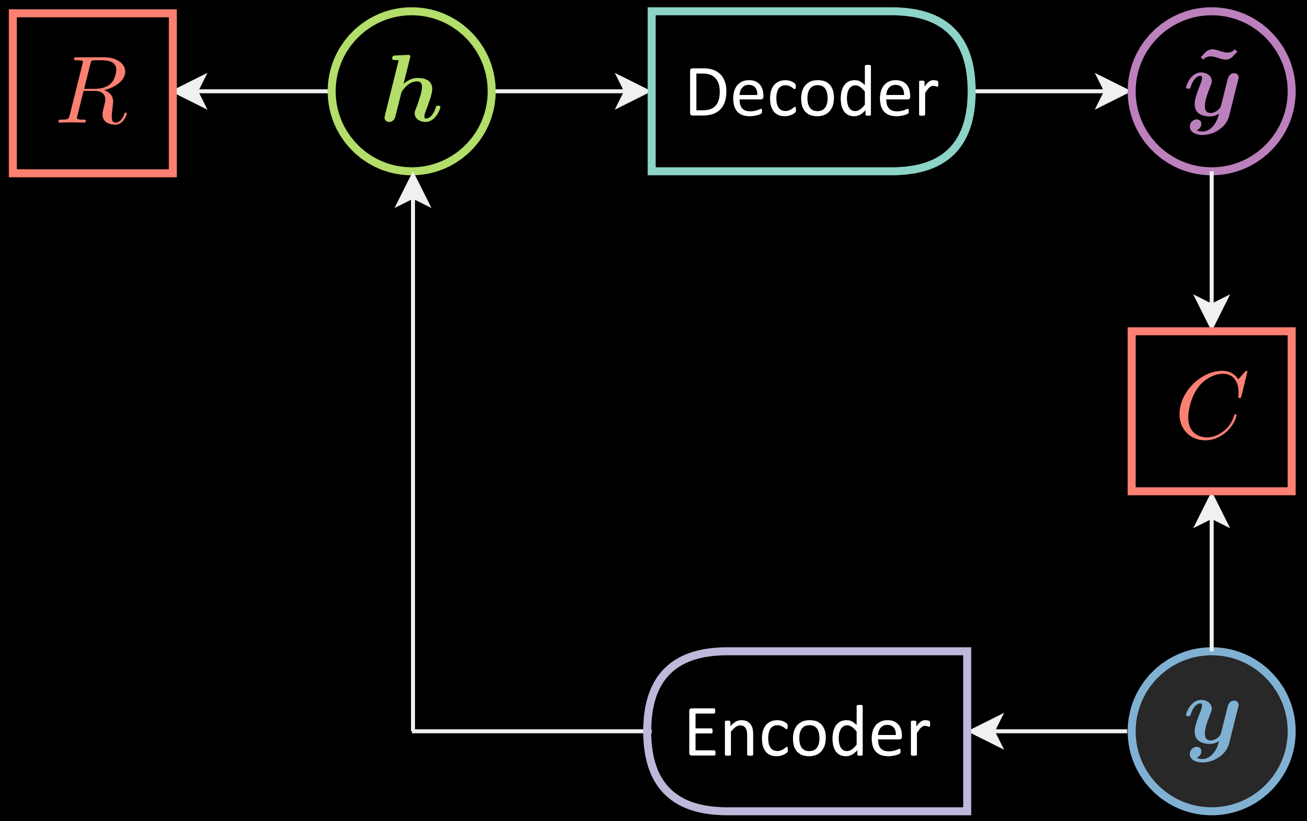
$$\tilde{\boldsymbol{y}} = g(\boldsymbol{W_y h} + \boldsymbol{b_y})$$

$$\boldsymbol{y}, \tilde{\boldsymbol{y}} \in \mathbb{R}^n$$

$$\boldsymbol{h} \in \mathbb{R}^d$$

$$\boldsymbol{W_h} \in \mathbb{R}^{d \times n}$$

$$\boldsymbol{W_y} \in \mathbb{R}^{n \times d}$$



$$\boldsymbol{h} = \mathrm{Enc}(\boldsymbol{y}) \quad \tilde{\boldsymbol{y}} = \mathrm{Dec}(\boldsymbol{h})$$

$$F(\boldsymbol{y}) = C(\boldsymbol{y}, \tilde{\boldsymbol{y}}) + R(\boldsymbol{h})$$

Reconstruction costs

real valued input

$$C(\boldsymbol{y}, \tilde{\boldsymbol{y}}) = \|\boldsymbol{y} - \tilde{\boldsymbol{y}}\|^2 = \|\boldsymbol{y} - \mathrm{Dec}\left[\mathrm{Enc}(\boldsymbol{y})\right]\|^2$$

binary input

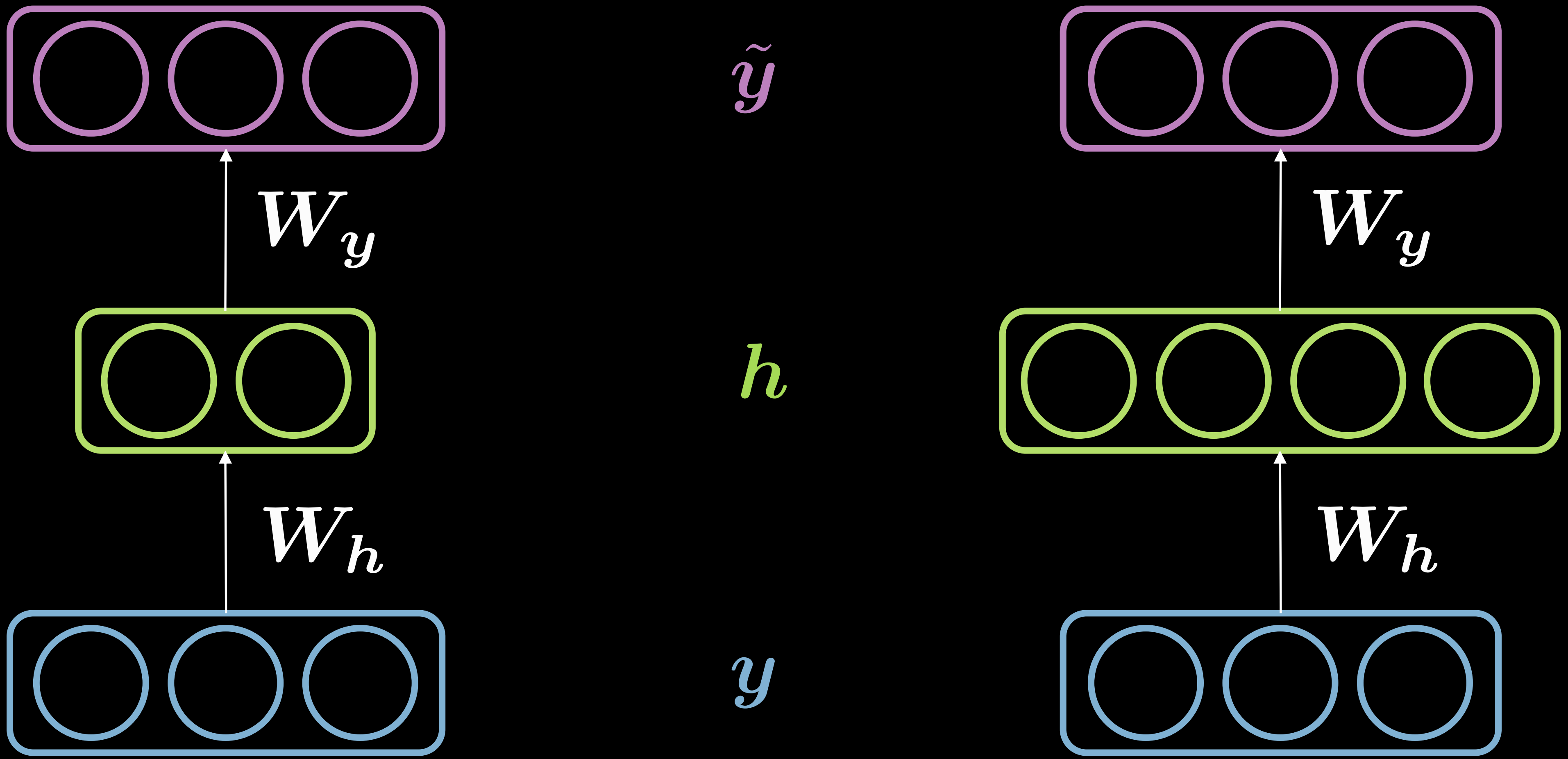
$$C(\boldsymbol{y}, \tilde{\boldsymbol{y}}) = -\sum_{i=1}^{n}\left[y_i \log(\tilde{y}_i) + (1 - y_i)\log(1 - \tilde{y}_i)\right]$$

Loss functional

$$\mathcal{L}(F(\cdot), \boldsymbol{Y}) = \frac{1}{m} \sum_{j=1}^{m} \ell(F(\cdot), \boldsymbol{y}^{(j)}) \in \mathbb{R}$$

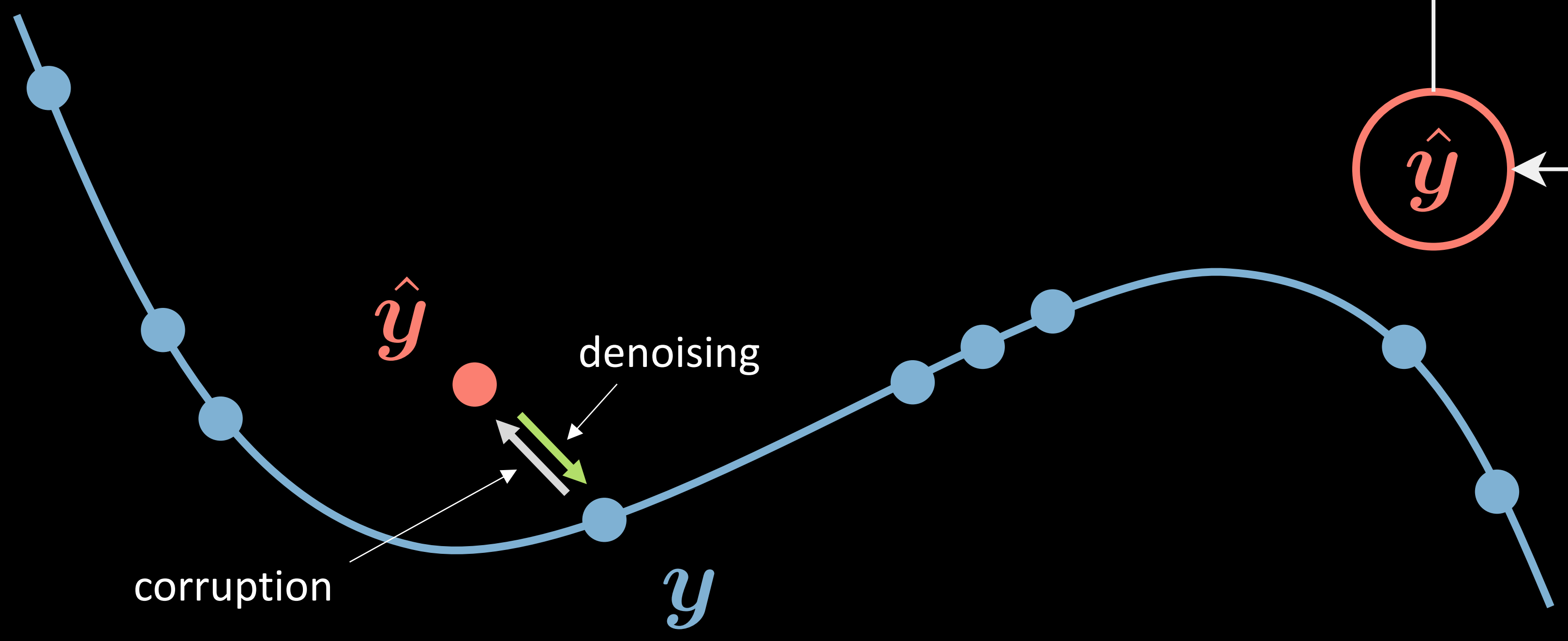
$$\ell_{\text{energy}}(F(\cdot), \boldsymbol{y}) = F(\boldsymbol{y})$$

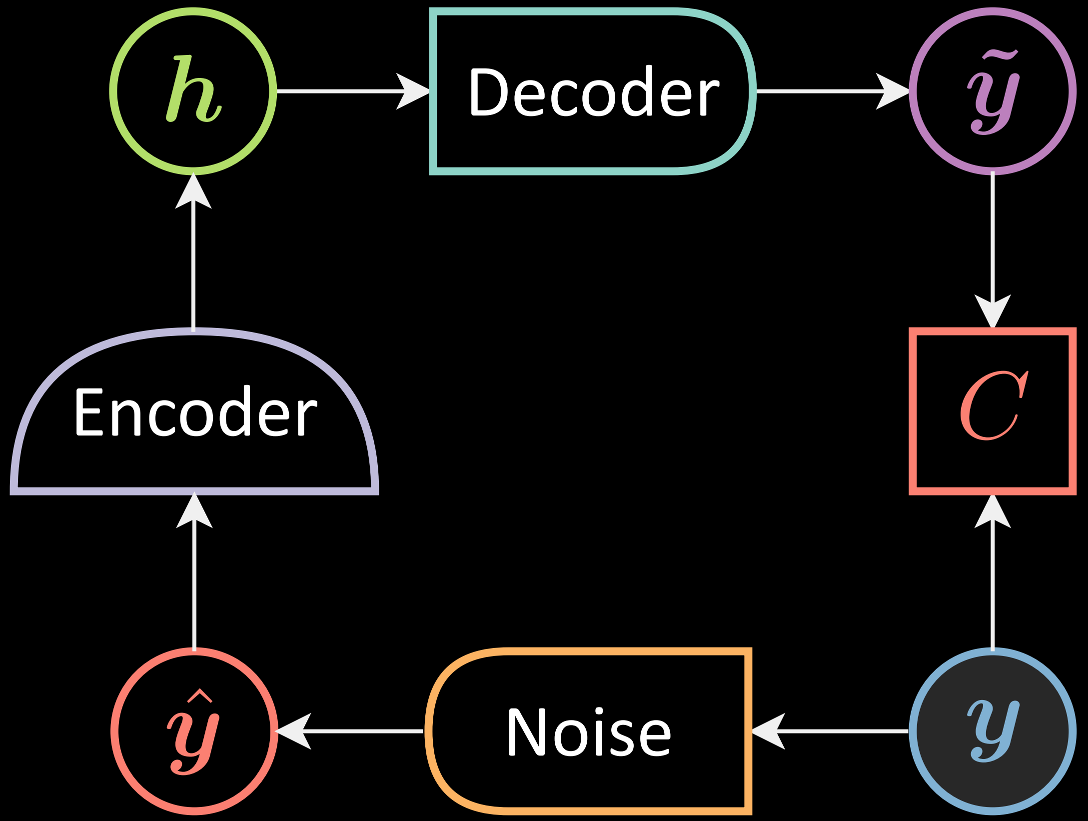
Under-/over-complete hidden layer

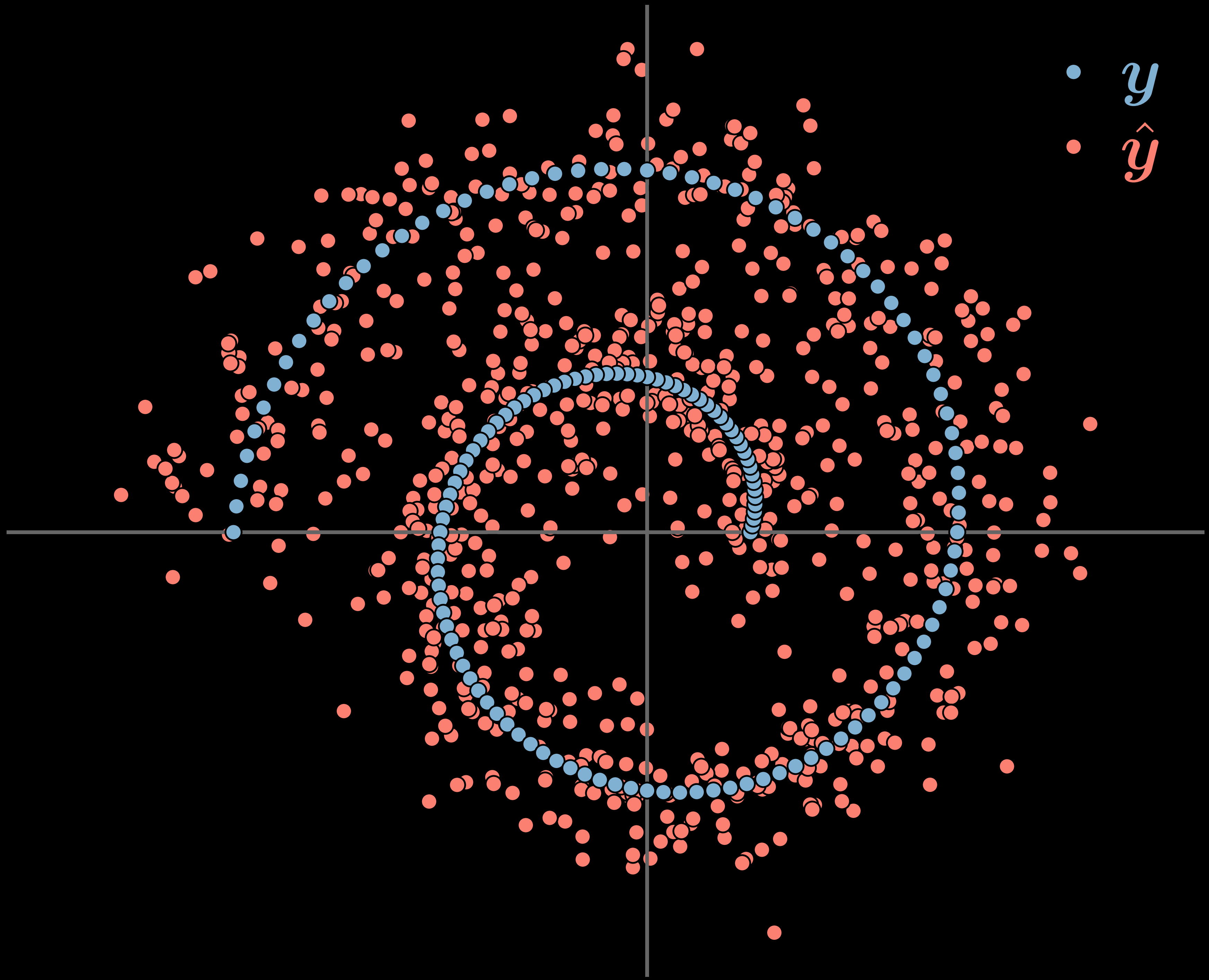


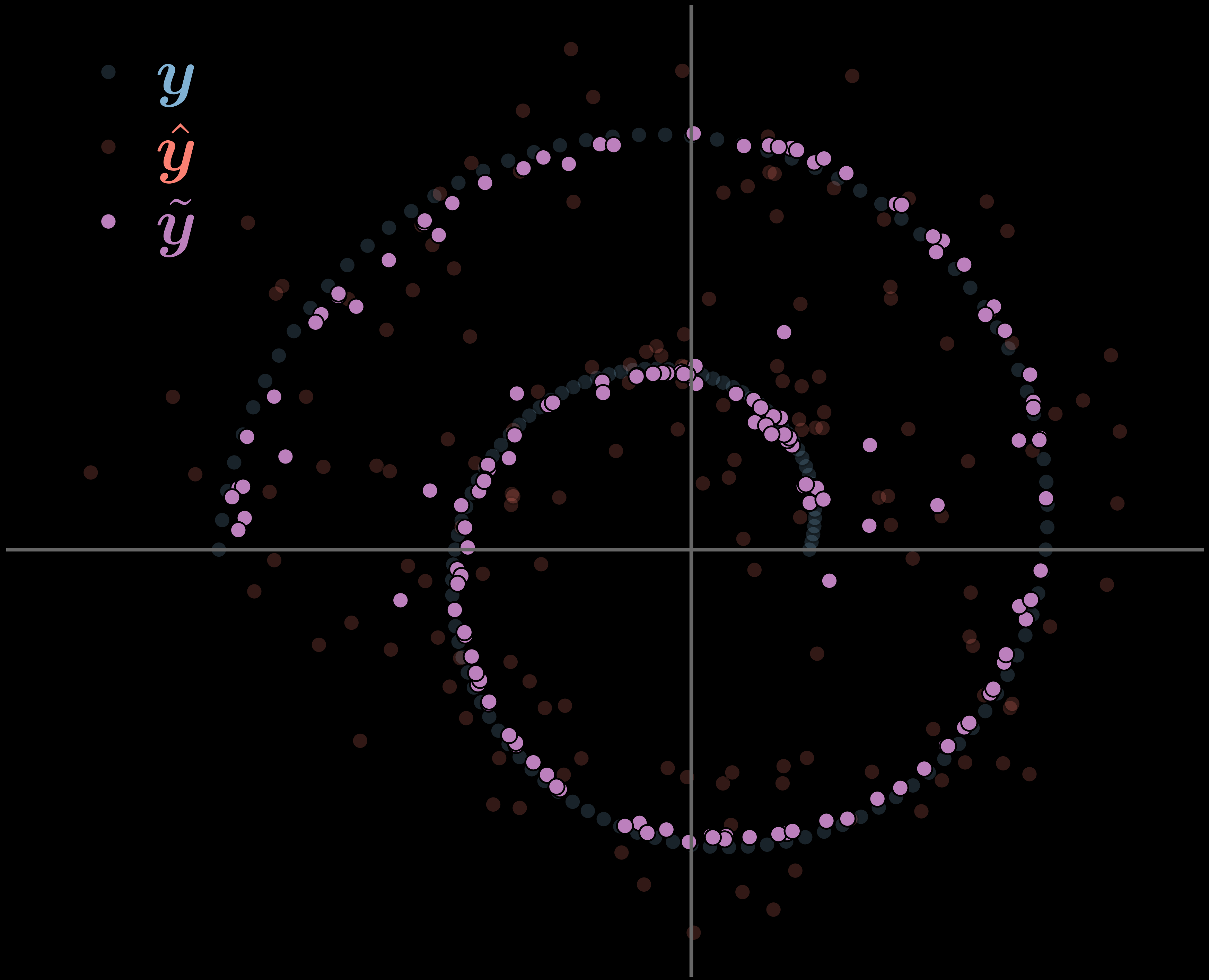
Denoising autoencoder

$$\hat{y} \sim p(\hat{y} \mid y)$$

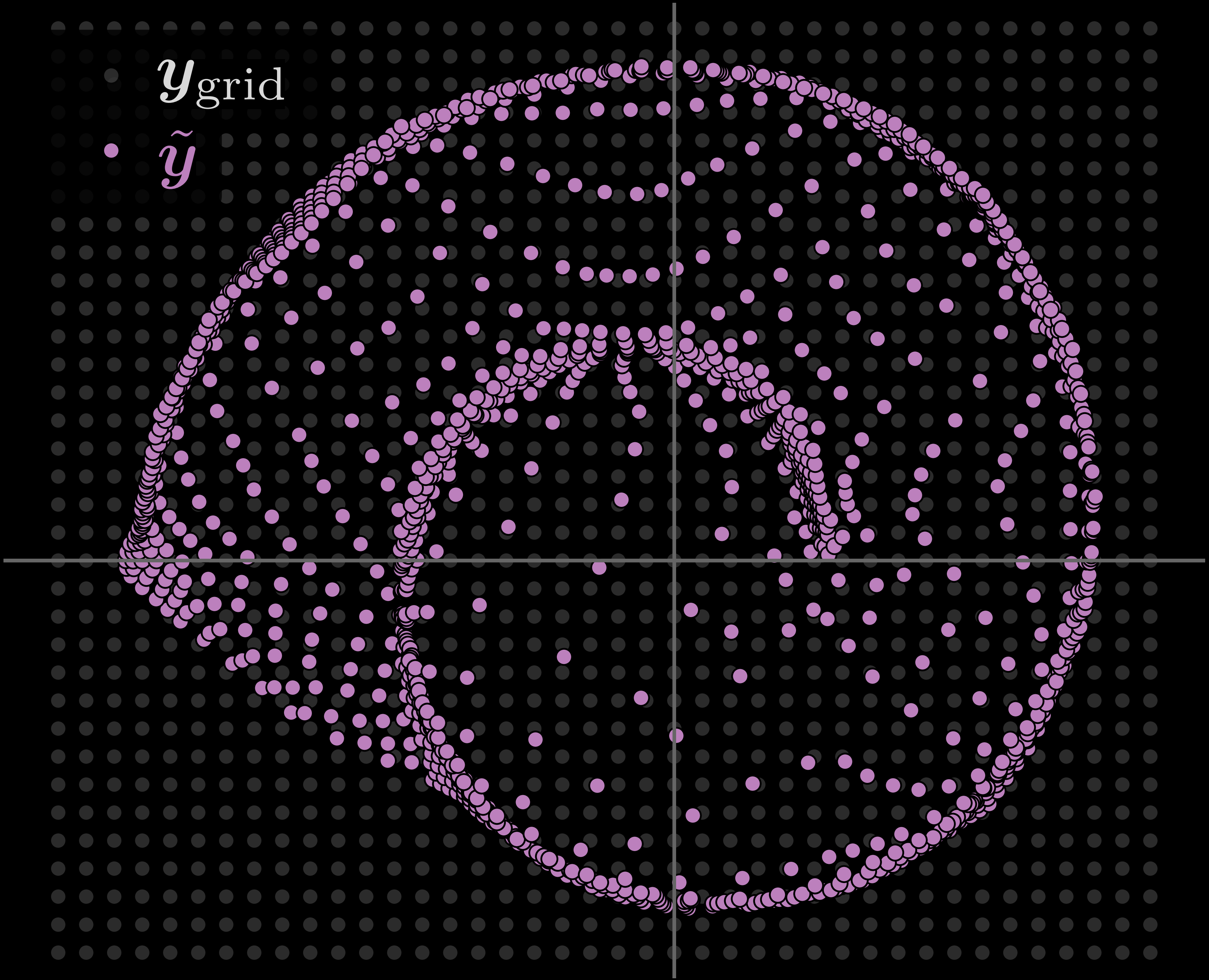




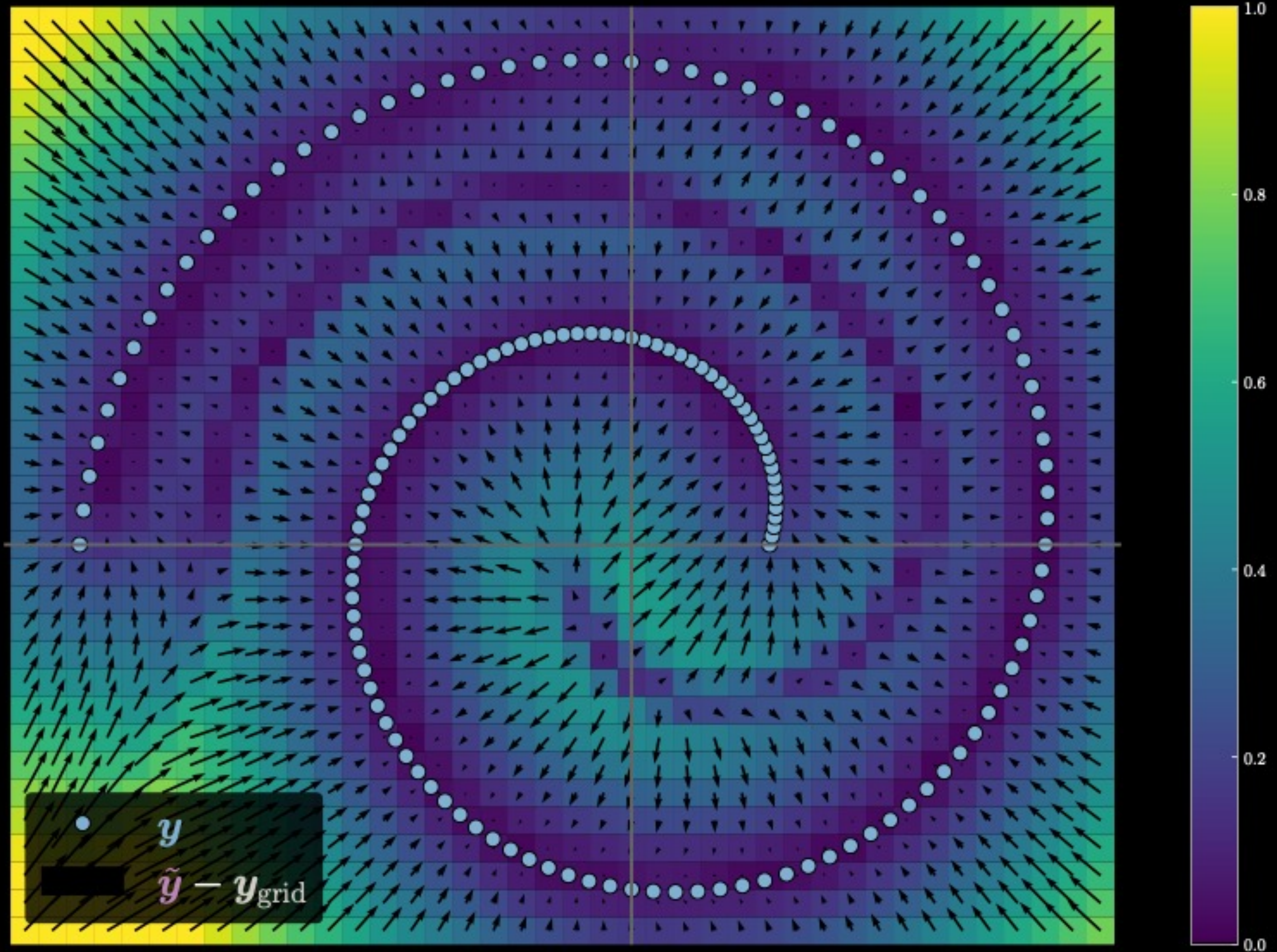

y
$\hat{y}$



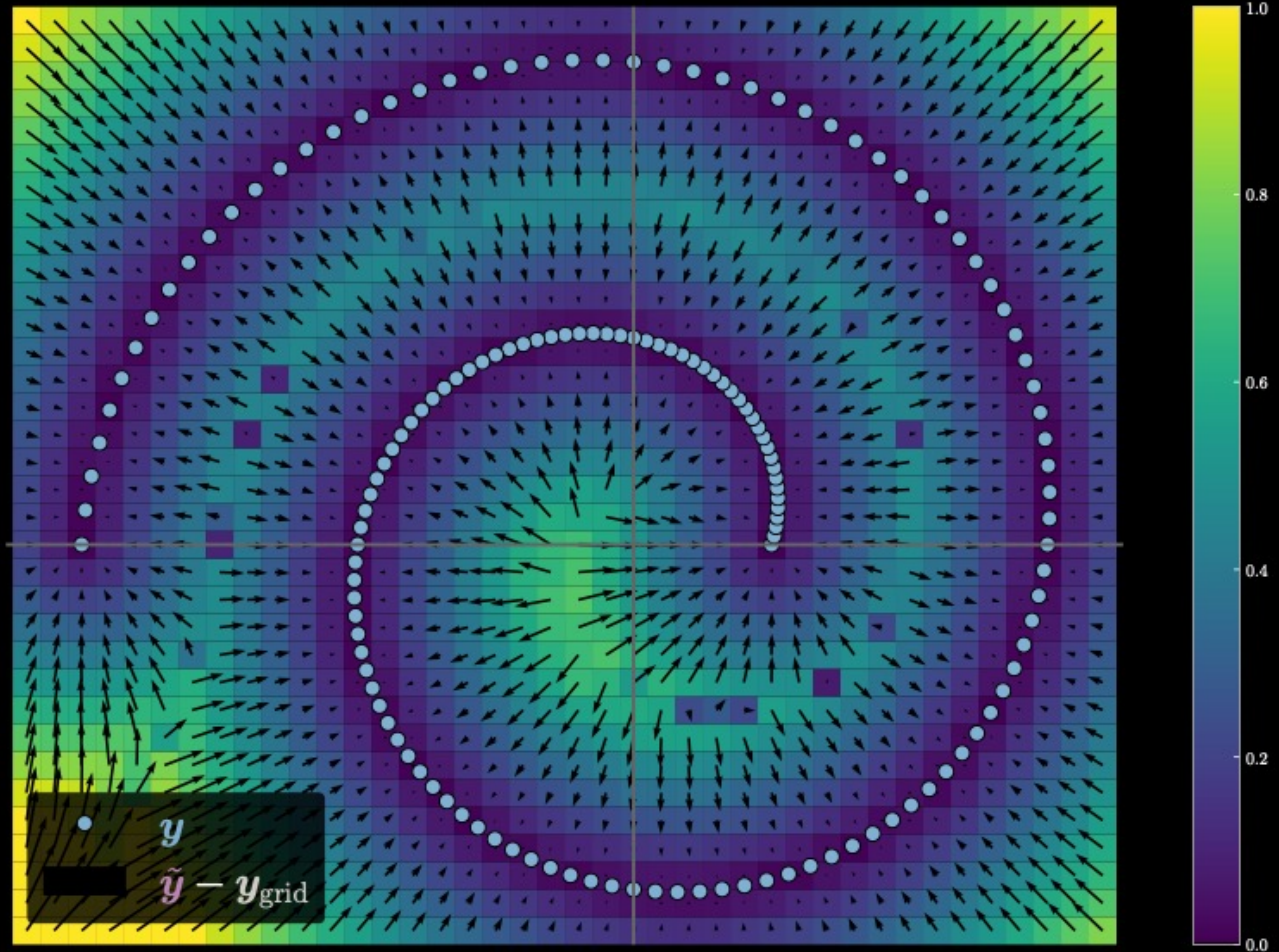
y
$\hat{y}$
$\tilde{y}$


y_{grid}
$\tilde{y}$

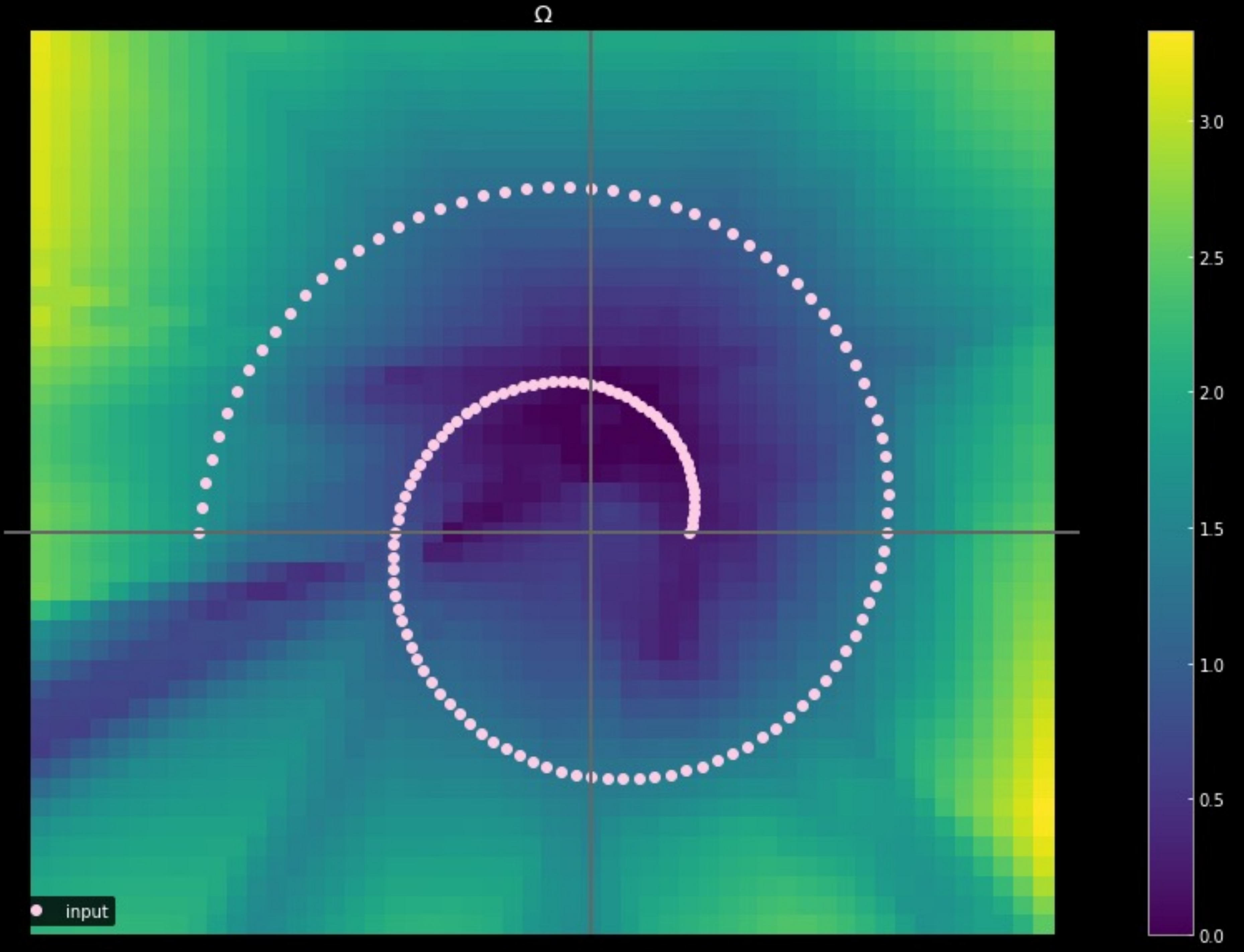
Denoising AE



Denoising AE, near. neigh.



Sparse autoencoder



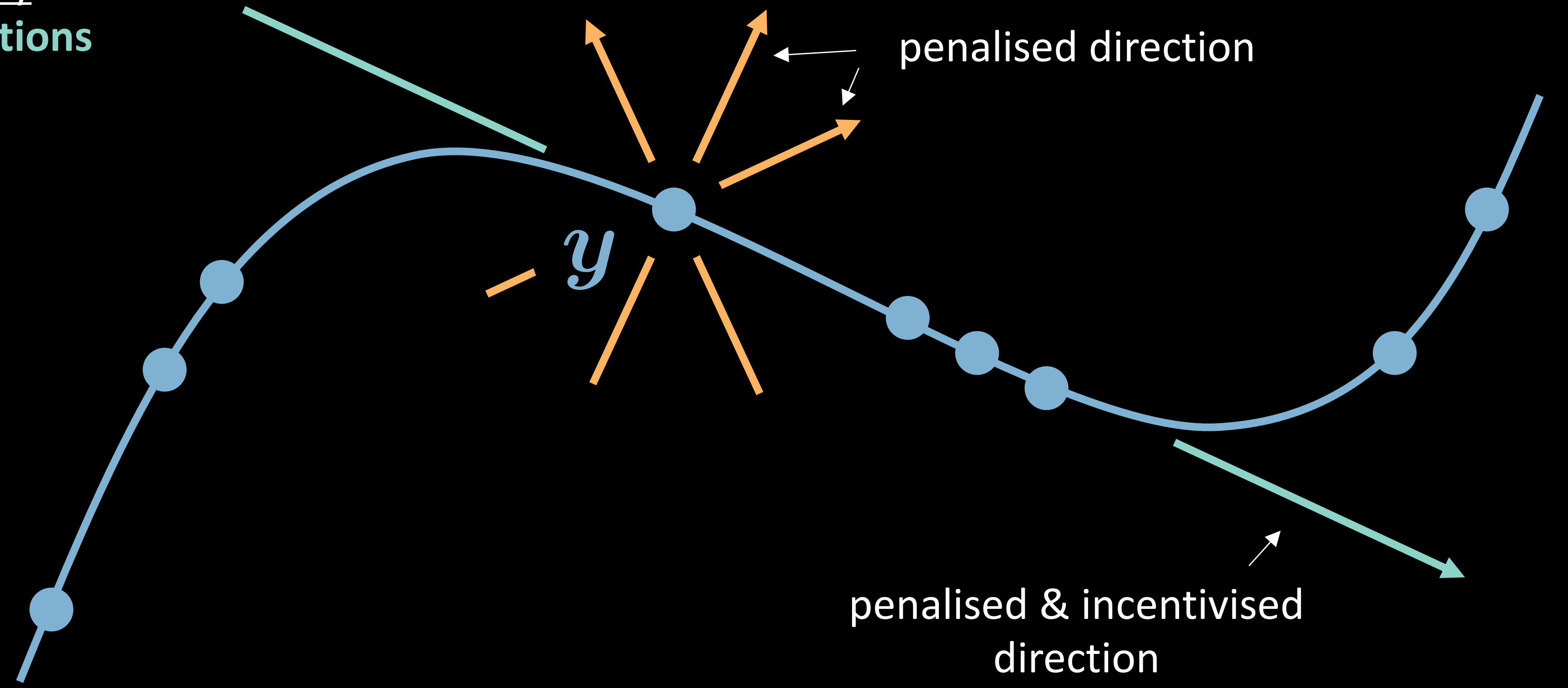
Contractive autoencoder

penalises <u>sensitivity</u>
to the **any direction**

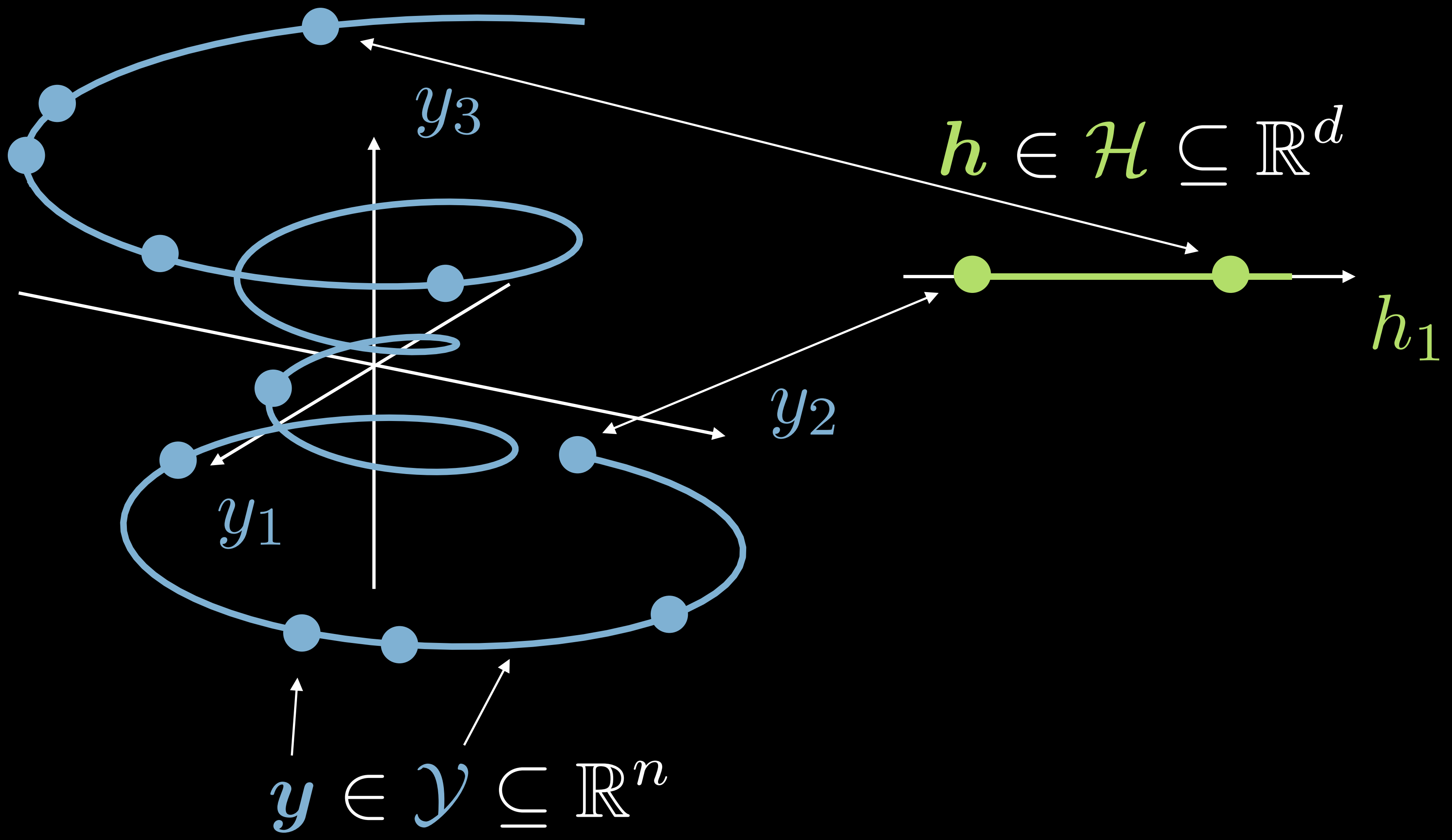
$$F(y) = \underbrace{C(y, \tilde{y})} + R(h)$$

$$R(h) = \lambda \|\nabla_y h\|^2$$

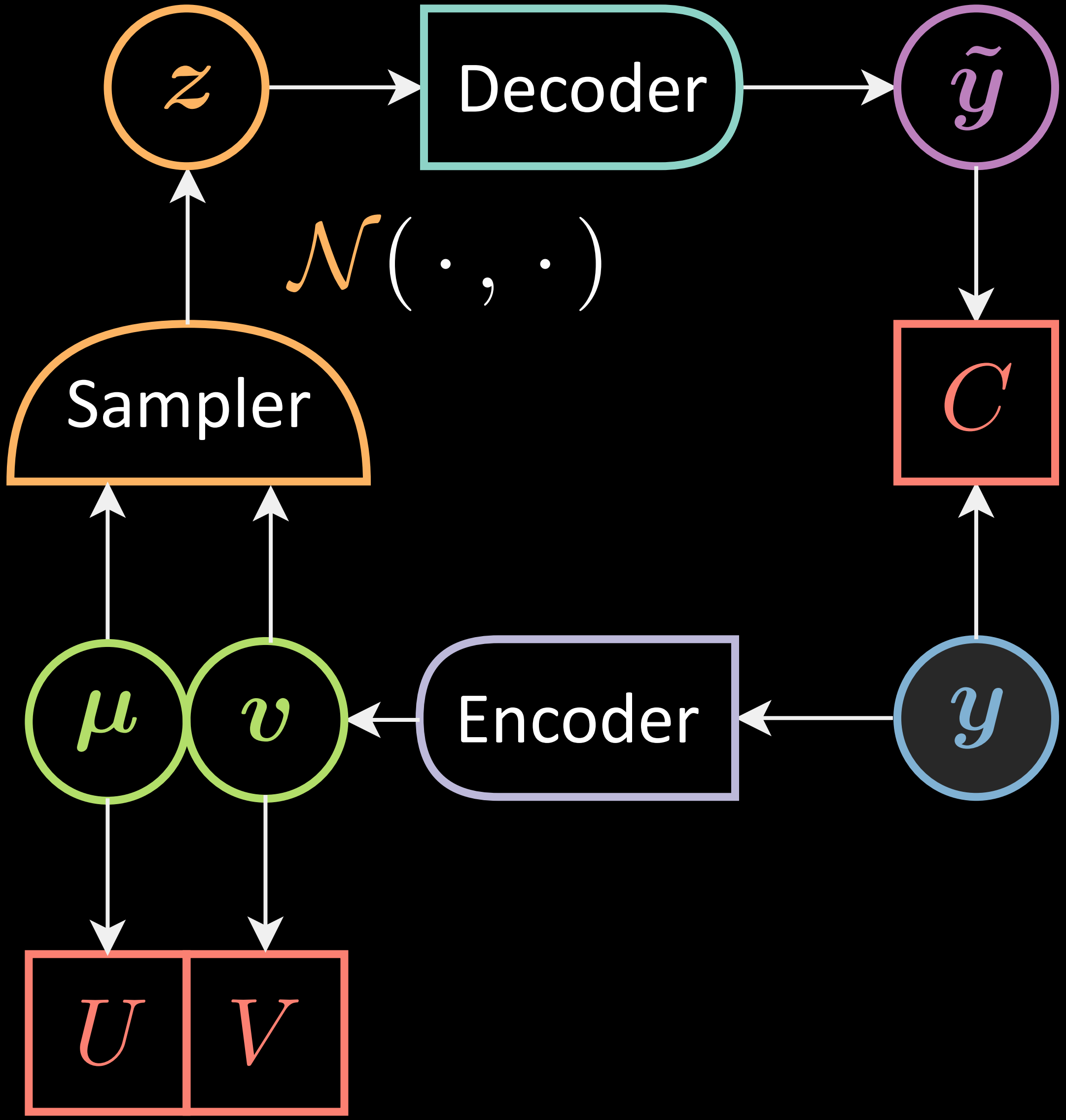
penalises <u>insensitivity</u> to
reconstruction directions

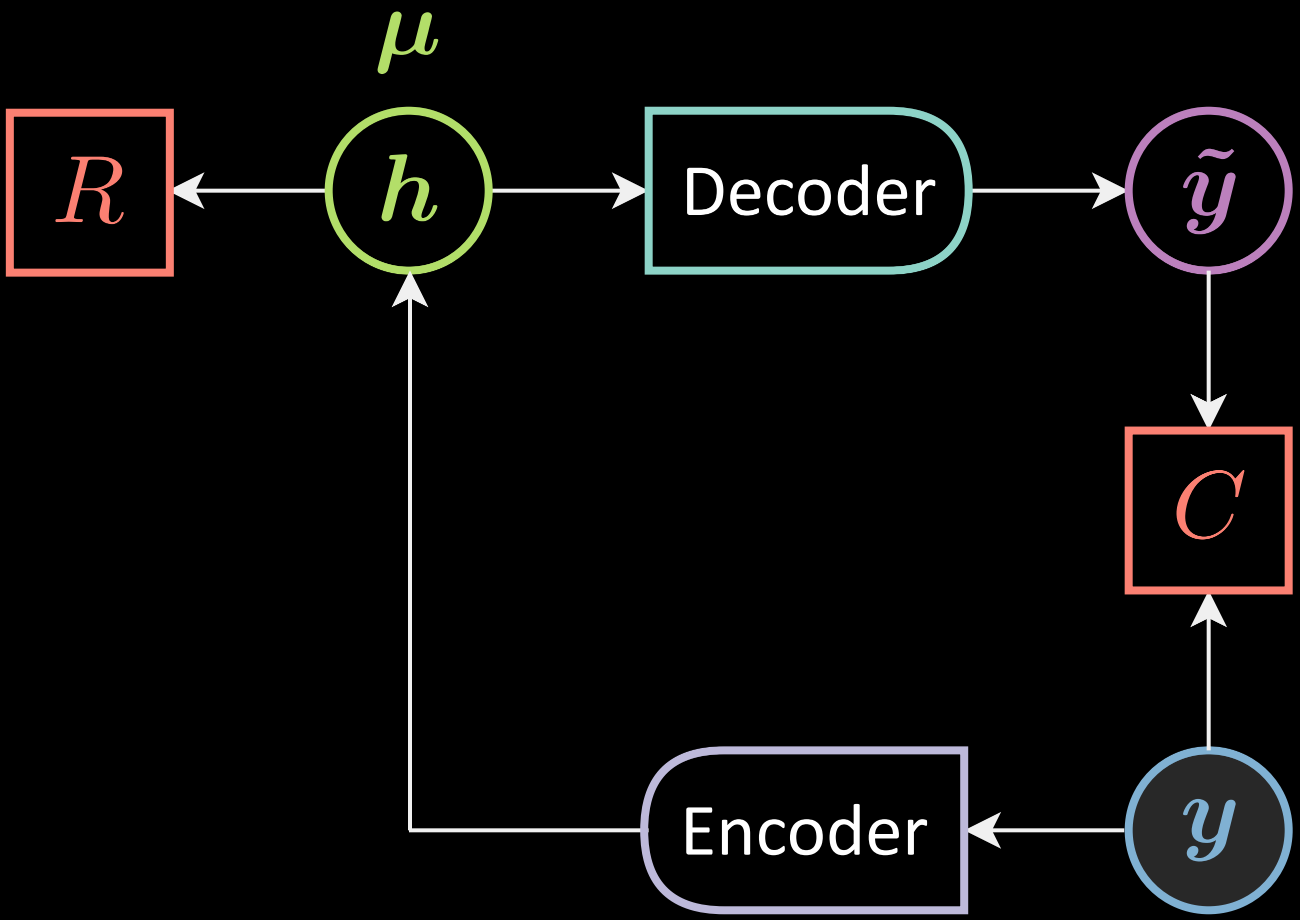


Autoencoder illustration



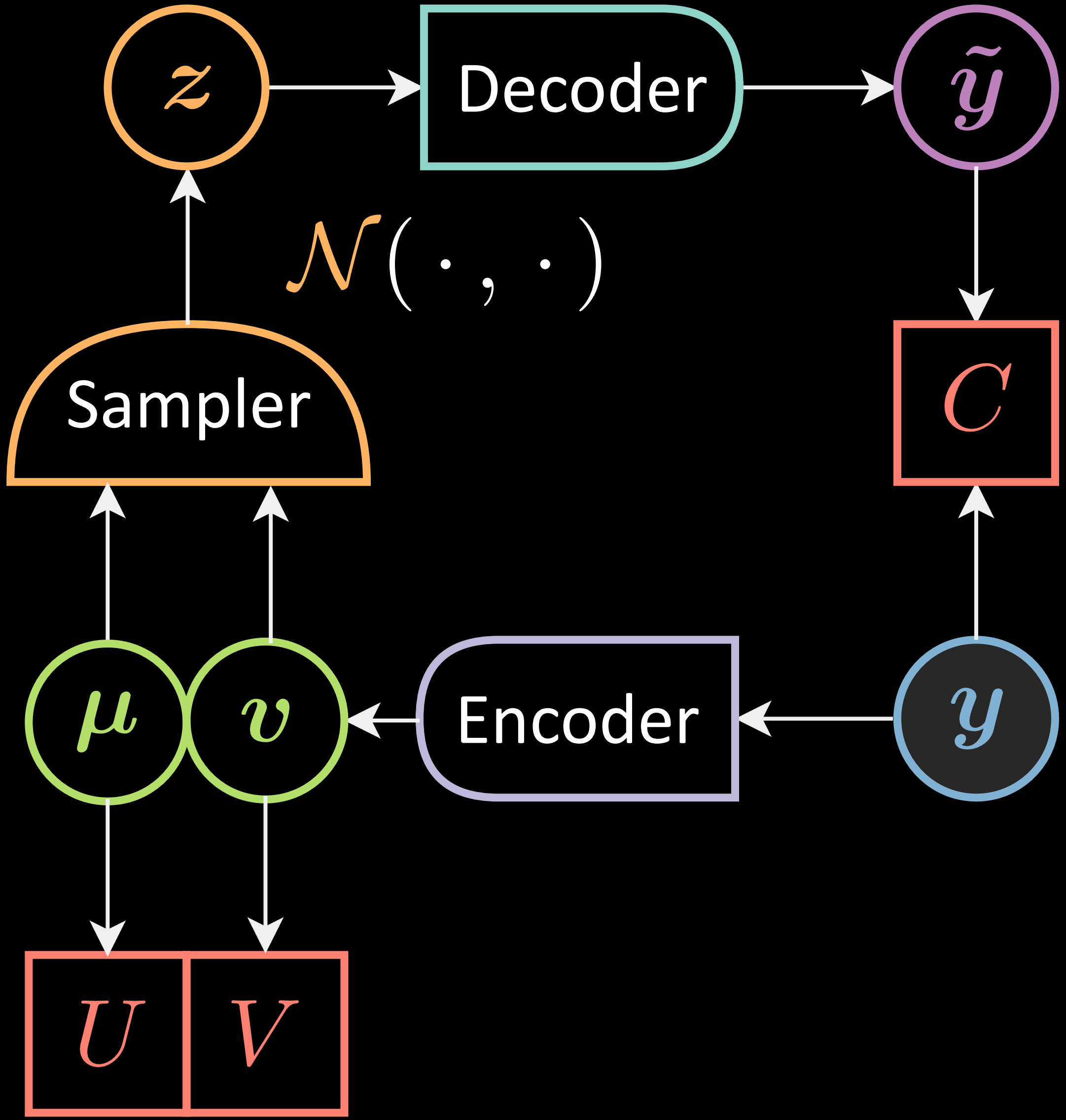
Variational AE

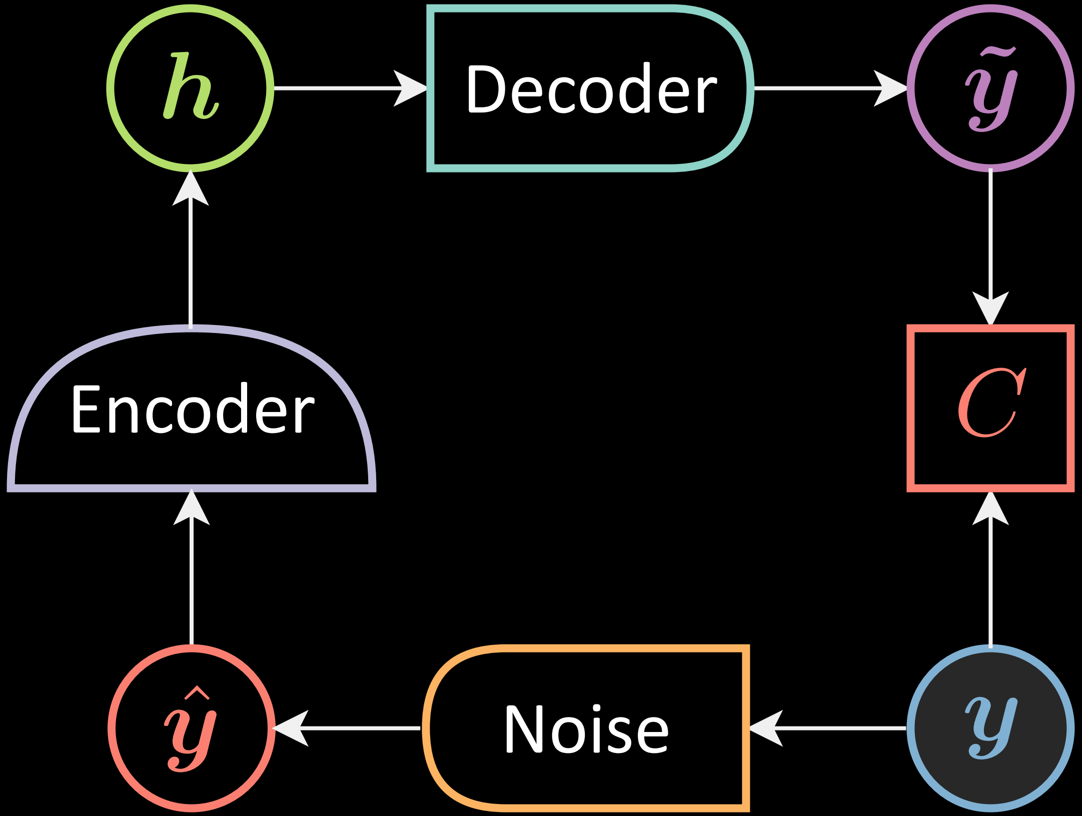




Basic autoencoder

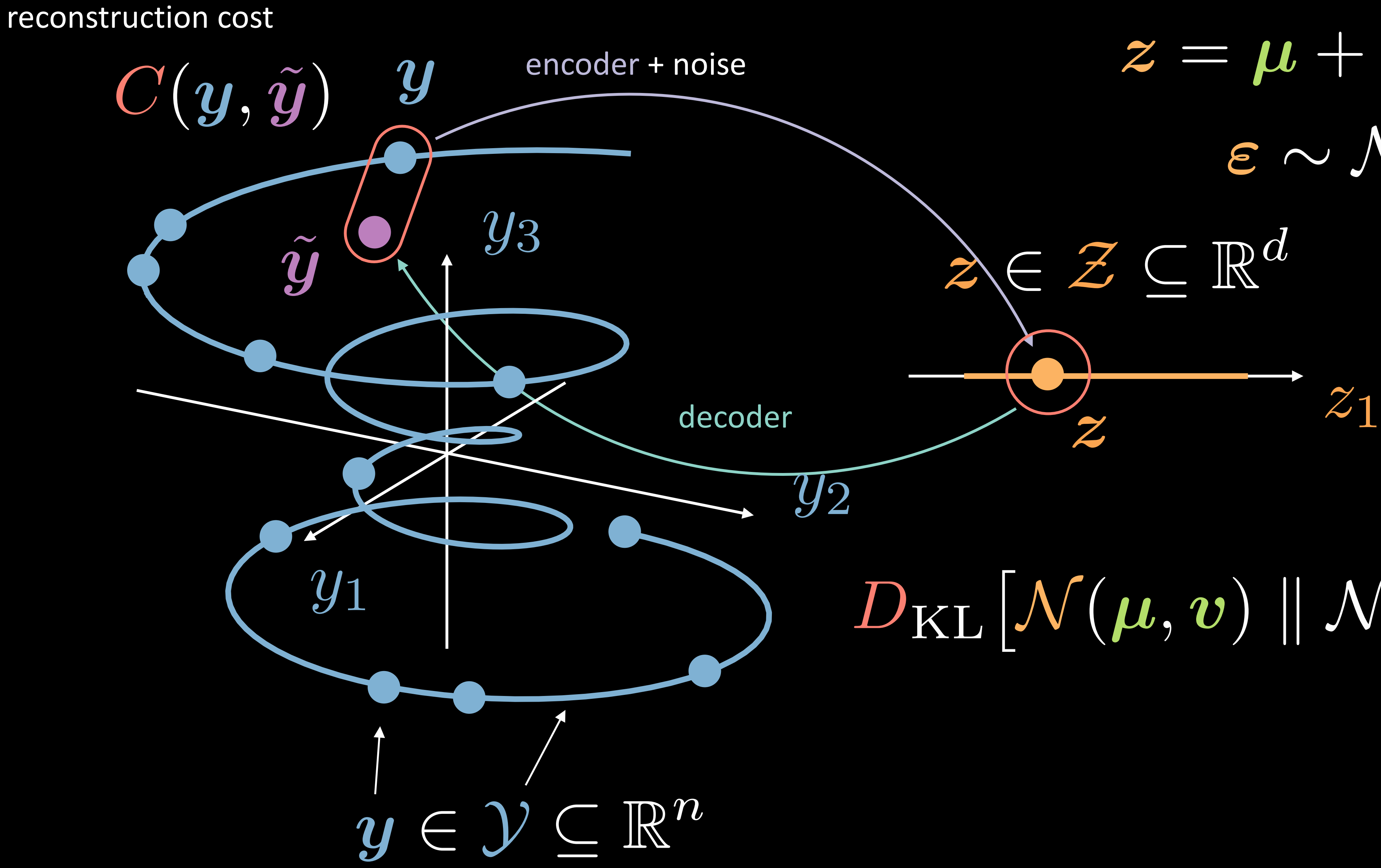
Variational AE





Denoising autoencoder

Variational autoencoder illustration



$$z = \mu + \varepsilon \odot \sqrt{v}$$

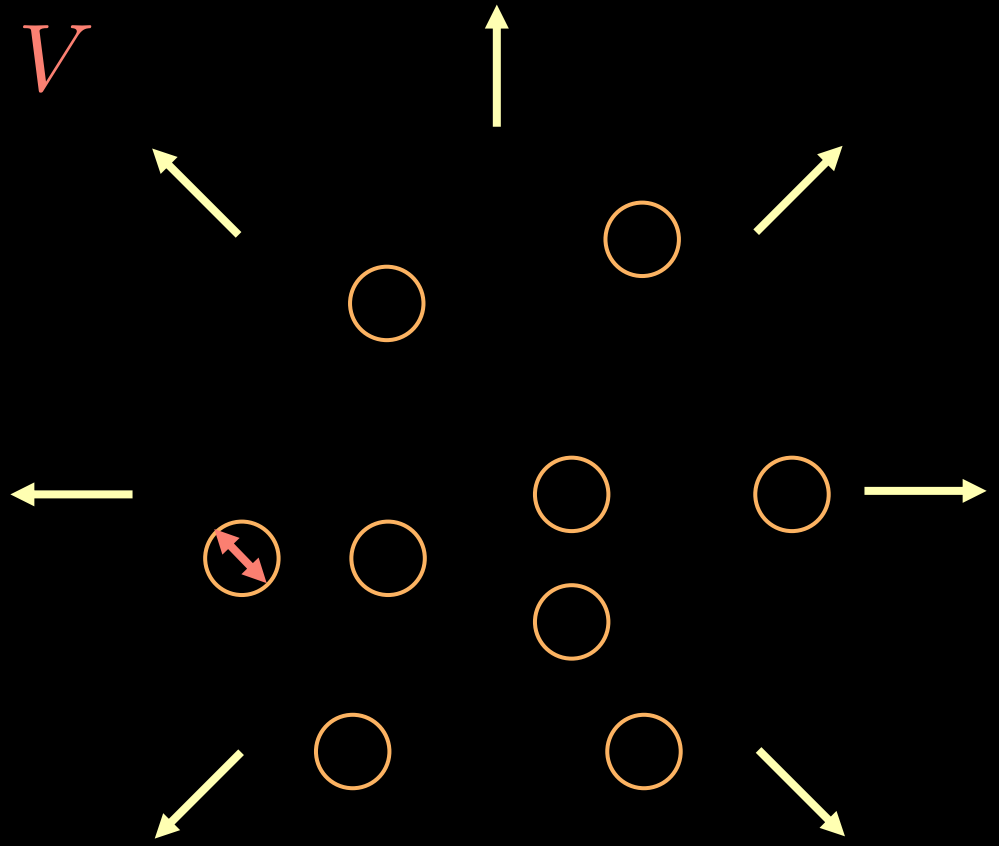
$$\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbb{I}_d)$$

$$D_{\mathrm{KL}}\left[\mathcal{N}(\mu, v) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{1})\right]$$

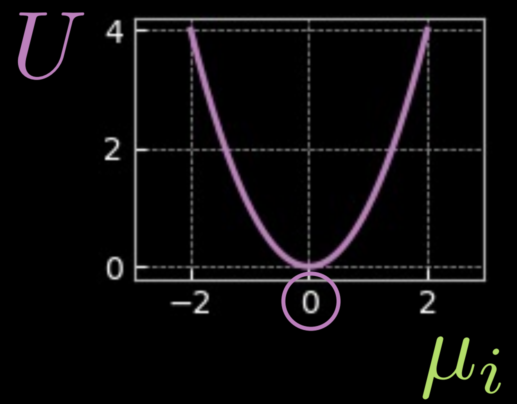
Variational auto-encoder

$$\tilde{F}(\boldsymbol{y}) = \boxed{C(\boldsymbol{y}, \tilde{\boldsymbol{y}})} + \beta D_{\mathrm{KL}}\left[\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{v}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{1})\right]$$

$$= C(\boldsymbol{y}, \tilde{\boldsymbol{y}}) + \frac{\beta}{2} \sum_{i=1}^{d} \underbrace{v_i - \log(v_i) - 1}_{V} + {\mu_i}^2$$

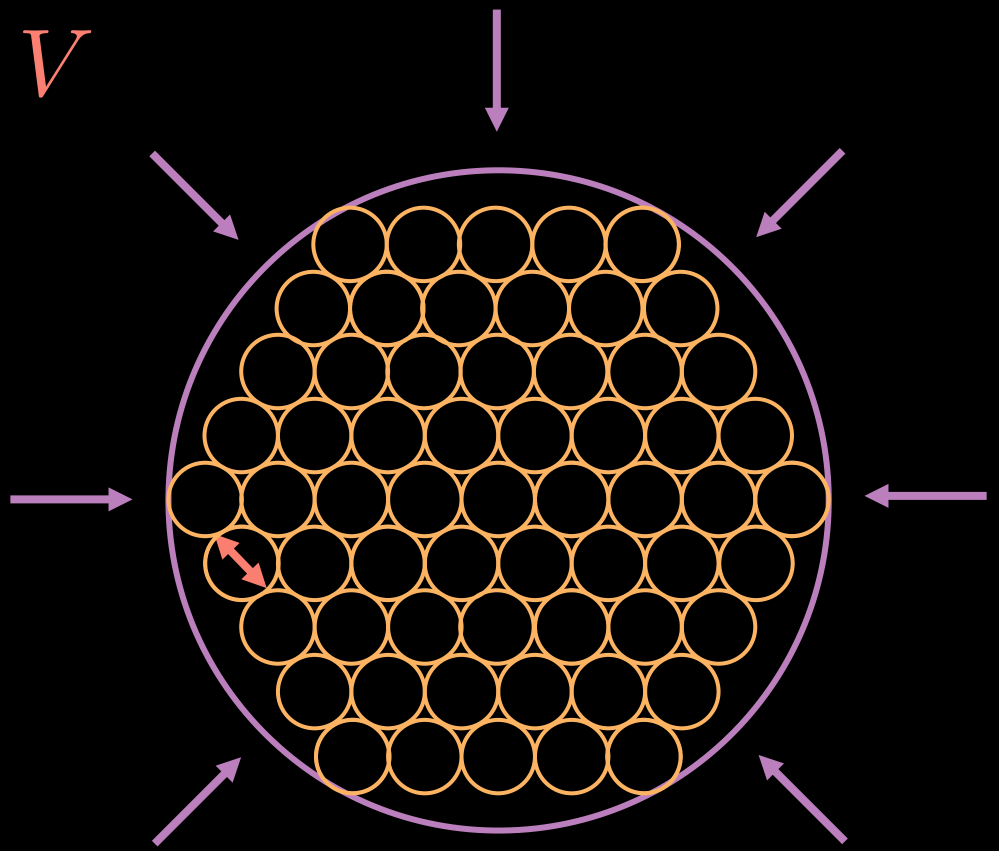


Variational auto-encoder



$$\tilde{F}(\boldsymbol{y}) = C(\boldsymbol{y},\tilde{\boldsymbol{y}}) + \beta D_{\mathrm{KL}}\left[\mathcal{N}(\boldsymbol{\mu},\boldsymbol{v}) \,\|\, \mathcal{N}(\boldsymbol{0},\boldsymbol{1})\right]$$

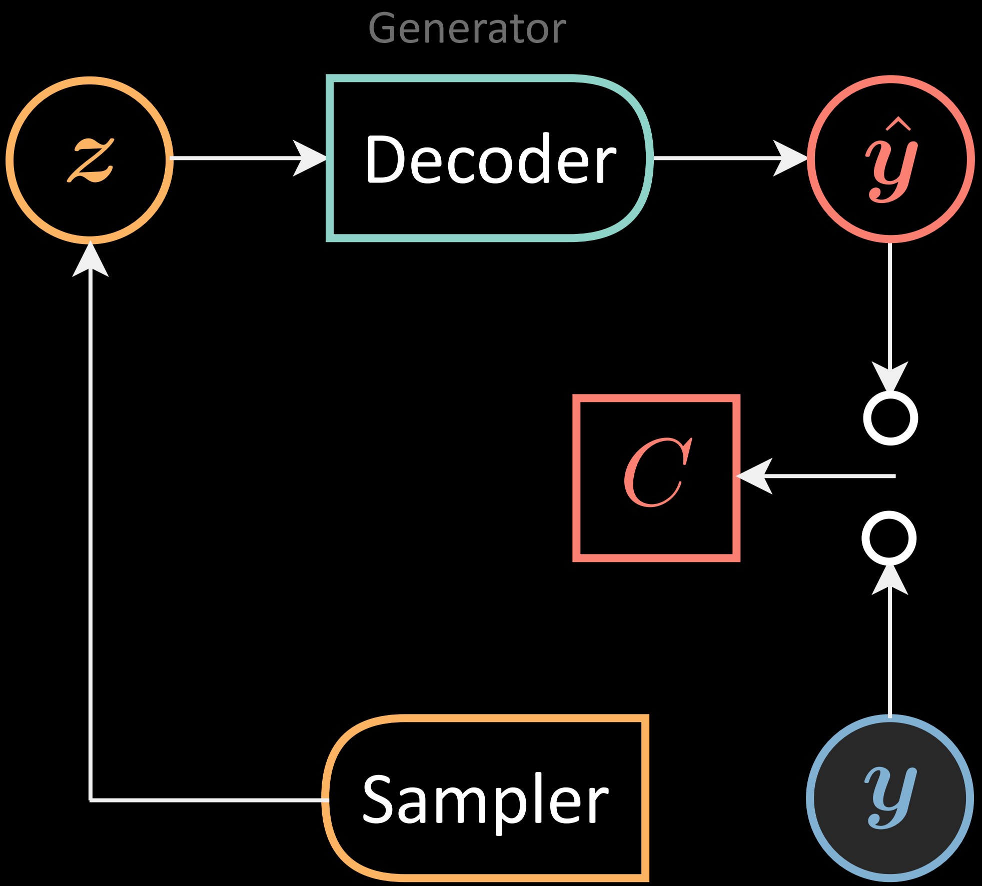
$$= C(\boldsymbol{y},\tilde{\boldsymbol{y}}) + \frac{\beta}{2}\sum_{i=1}^{d} \underbrace{v_i - \log(v_i) - 1}_{V} + \overbrace{{\mu_i}^2}^{U}$$

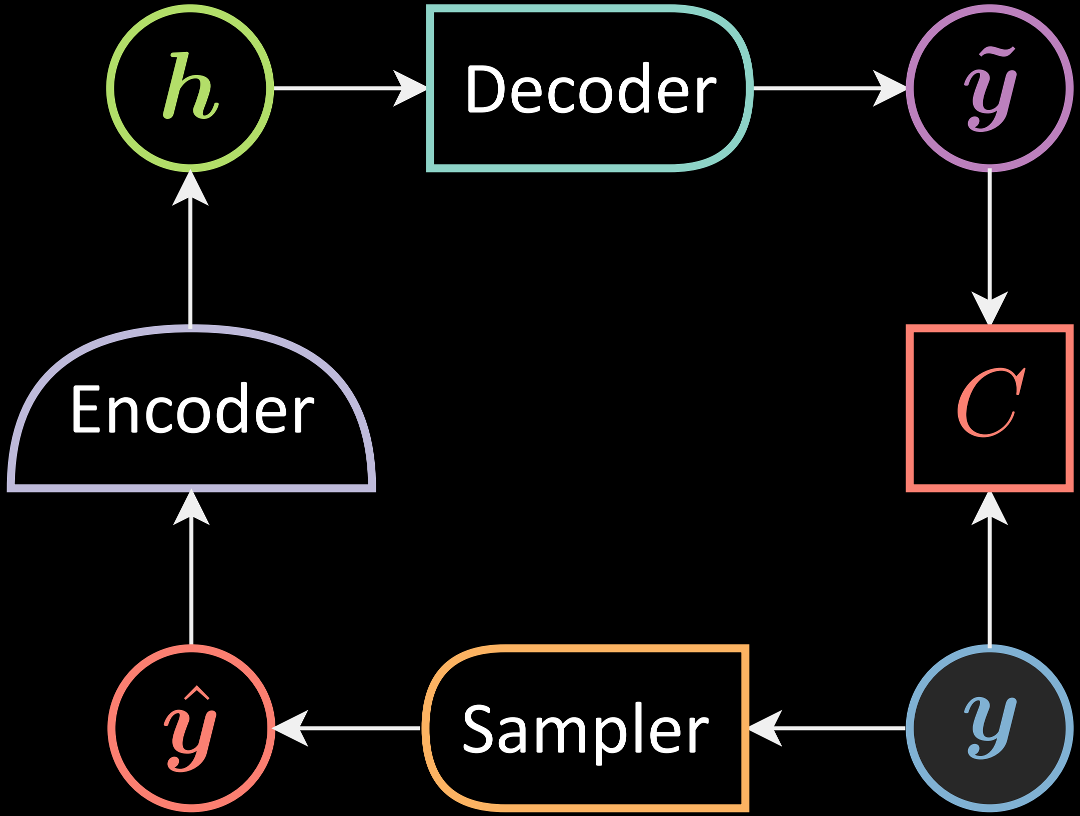


Generative adversarial nets

Unsupervised learning / Generative models

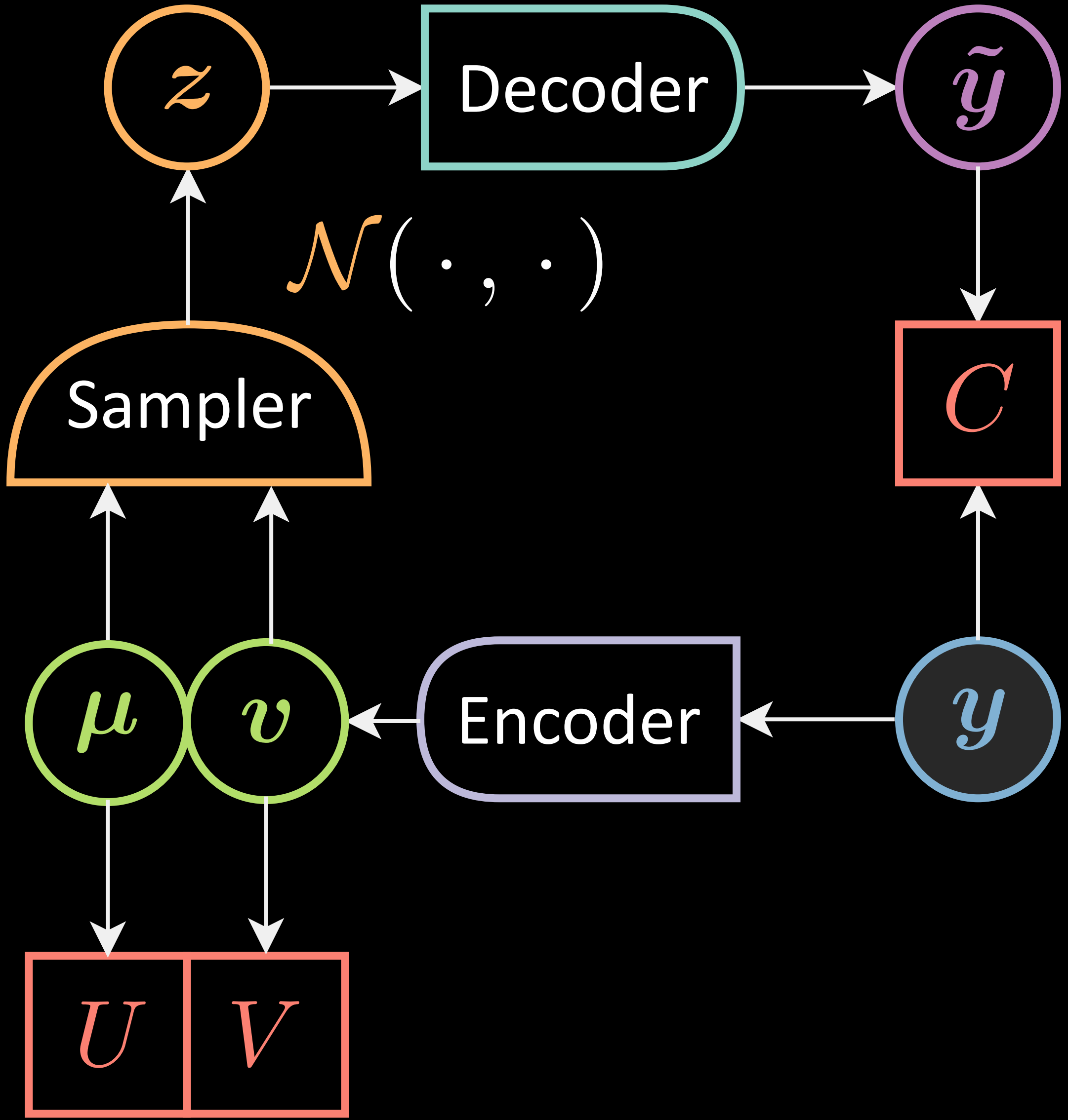
Gen$^{\underline{ve}}$ Adversarial Net

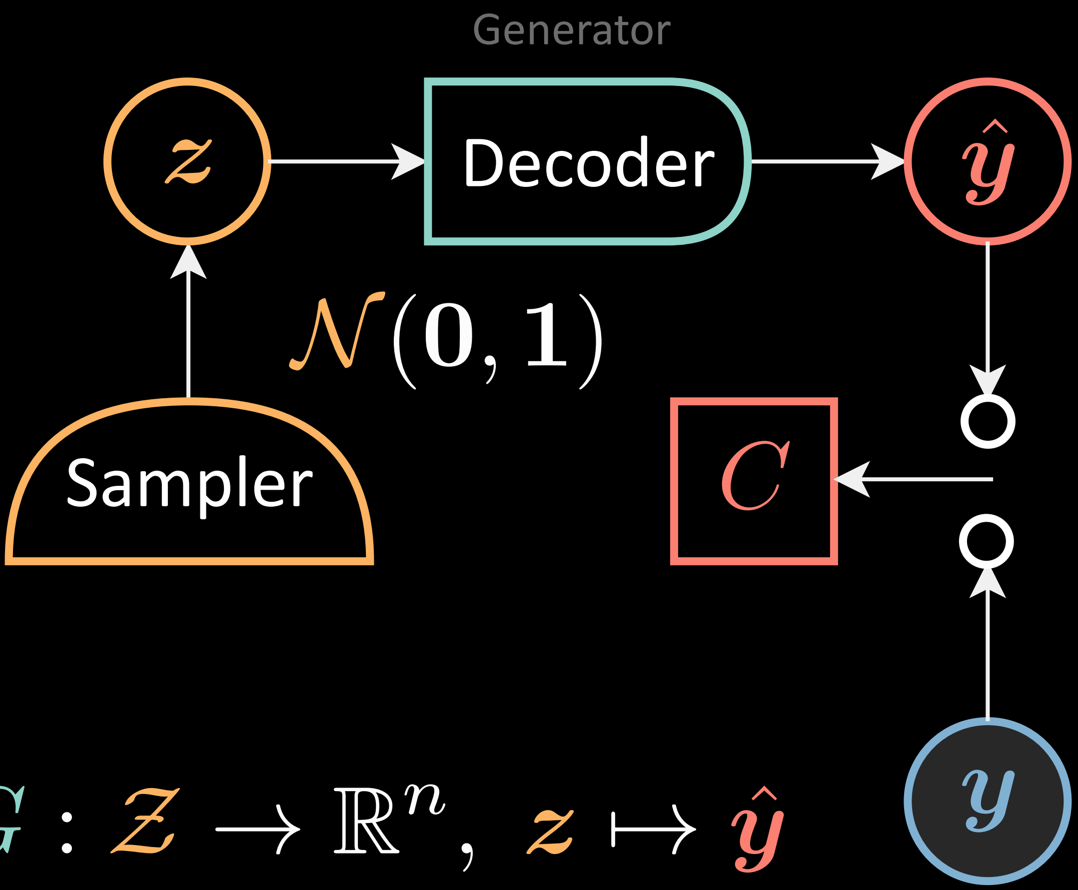




Denoising autoencoder

Variational AE





$$G : \mathcal{Z} \to \mathbb{R}^n, \quad z \mapsto \hat{y}$$

$$C : \mathbb{R}^n \to \mathbb{R}, \quad y \vee \hat{y} \mapsto c$$

Gen$^{\underline{\text{ve}}}$ Adversarial Net

Training

$$\ell_C(\boldsymbol{y}, \hat{\boldsymbol{y}}) = C(\boldsymbol{y}) + \left[m - C(\overbrace{\hat{\boldsymbol{y}}}^{G(\boldsymbol{z})})\right]^+$$

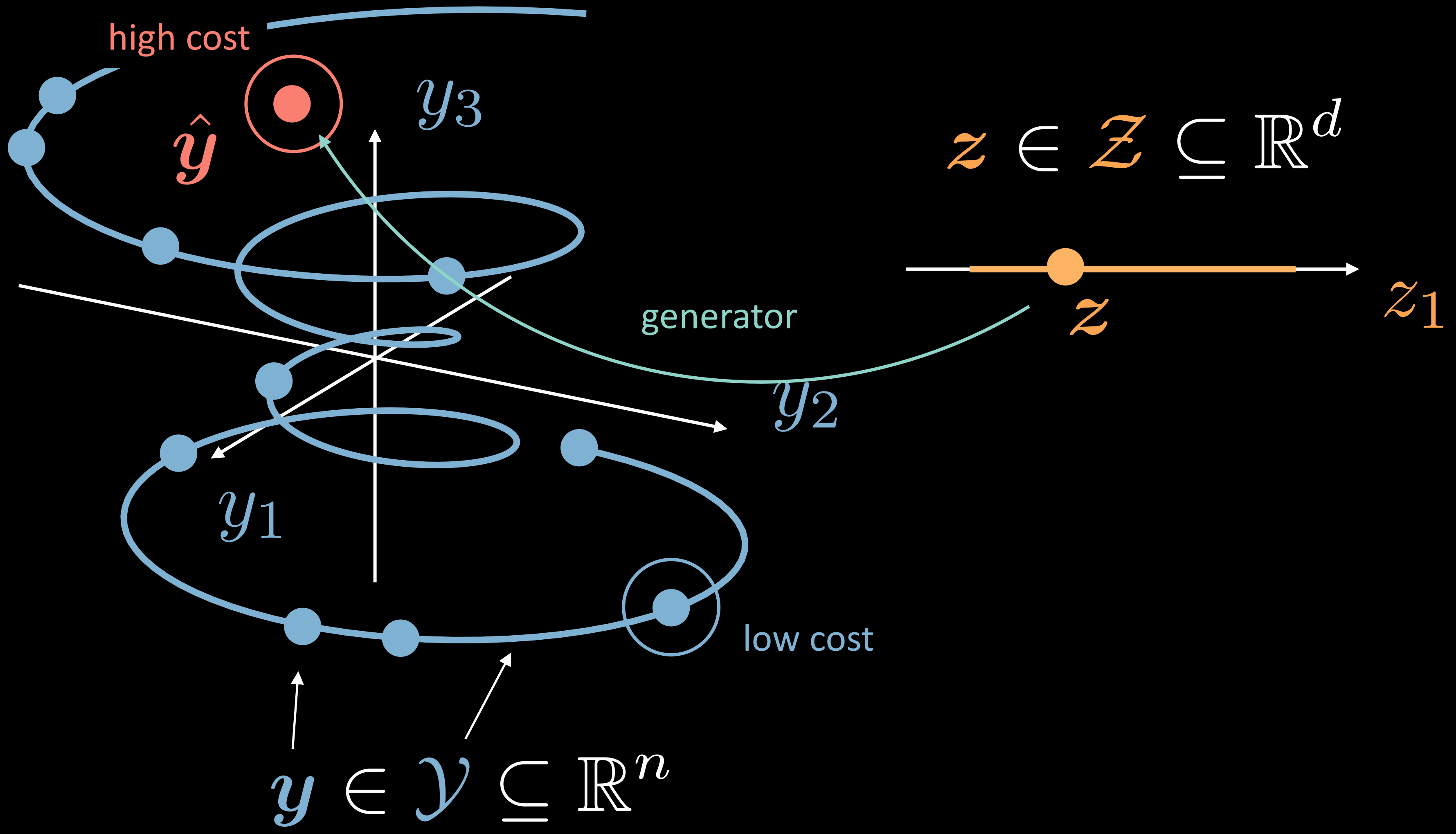
$$\ell_G(\boldsymbol{z}) = C\big(G(\boldsymbol{z})\big)$$

Possible choice of *C*(*y*):

$$C(\boldsymbol{y}) = \left\|\mathrm{Dec}\big(\mathrm{Enc}(\boldsymbol{y})\big) - \boldsymbol{y}\right\|^2$$

Zhao, Mathieu, LeCun (2016) Energy-based generative adversarial network

Generative adversarial network illustration



Major pitfalls

- Vanishing gradients
- Mode collapse
- Unstable convergence